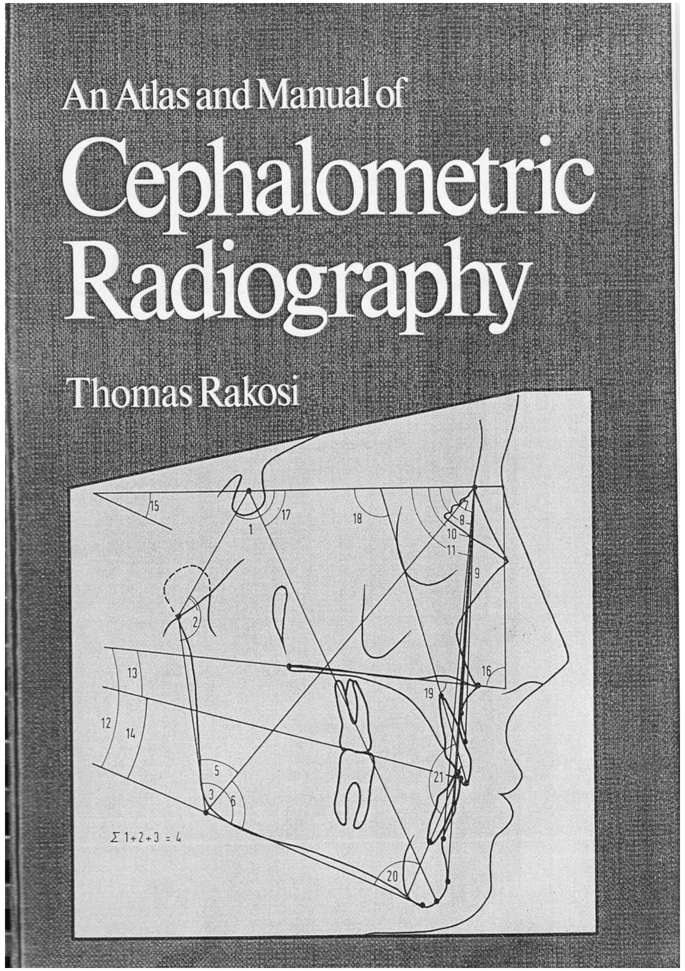
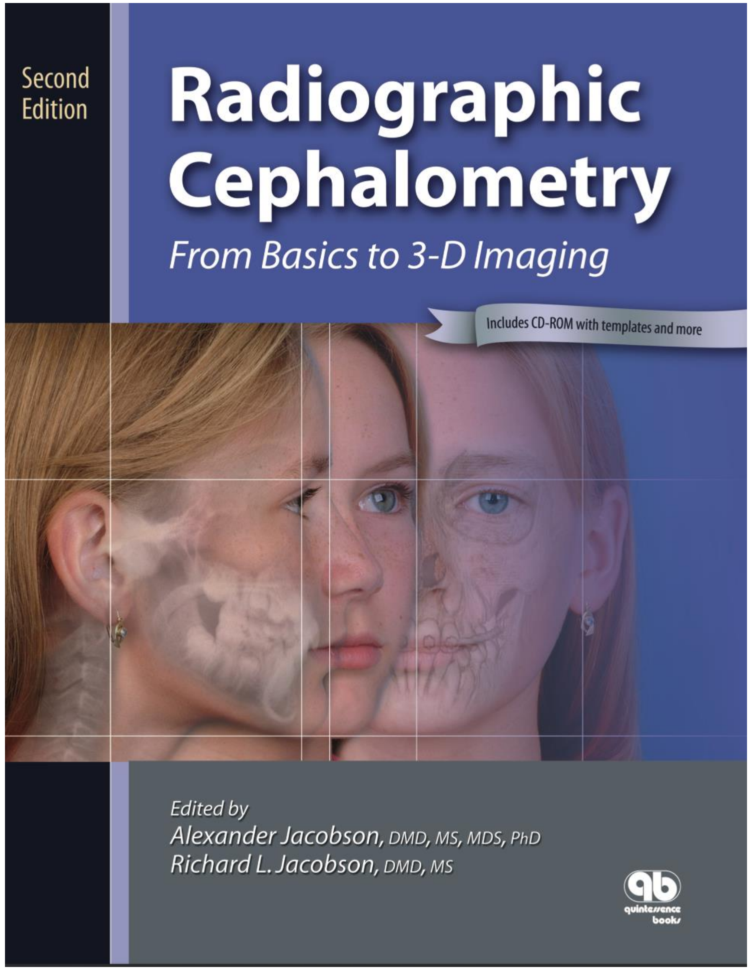
Cephalometry

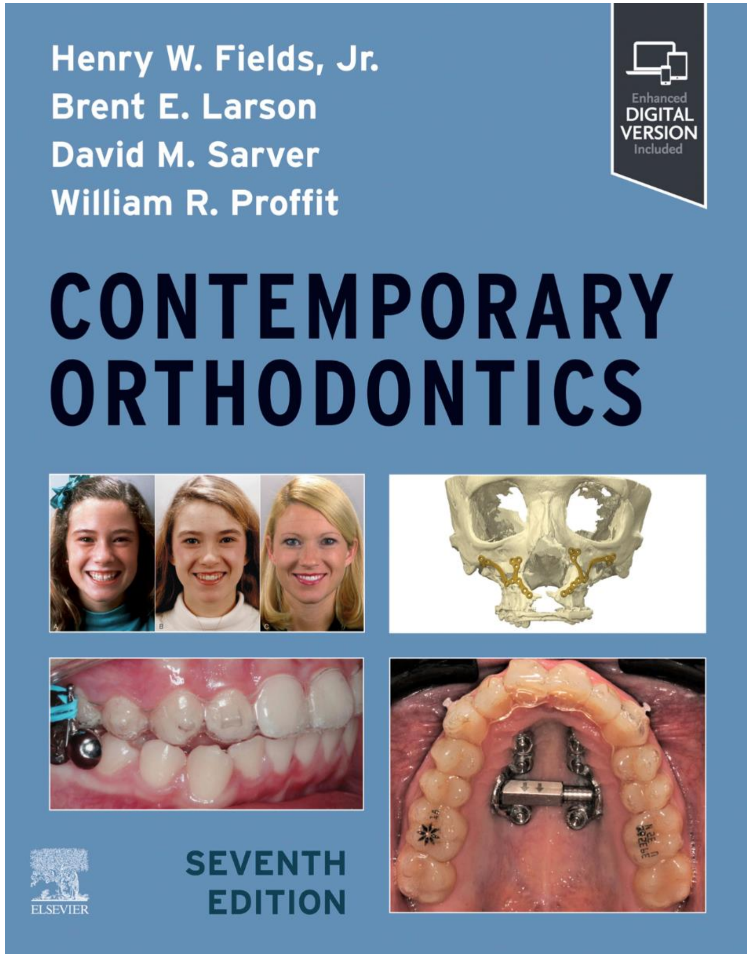
Mehdi Kashani DDS MSc

References:

ABO American Board of Orthodontics

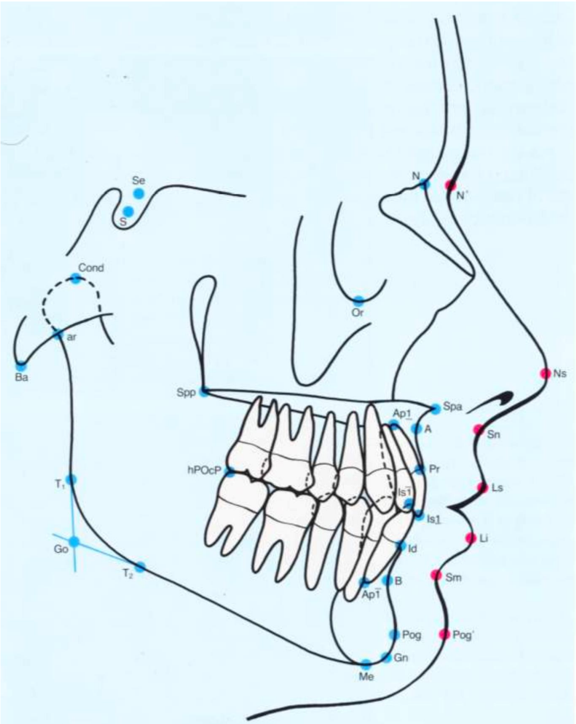


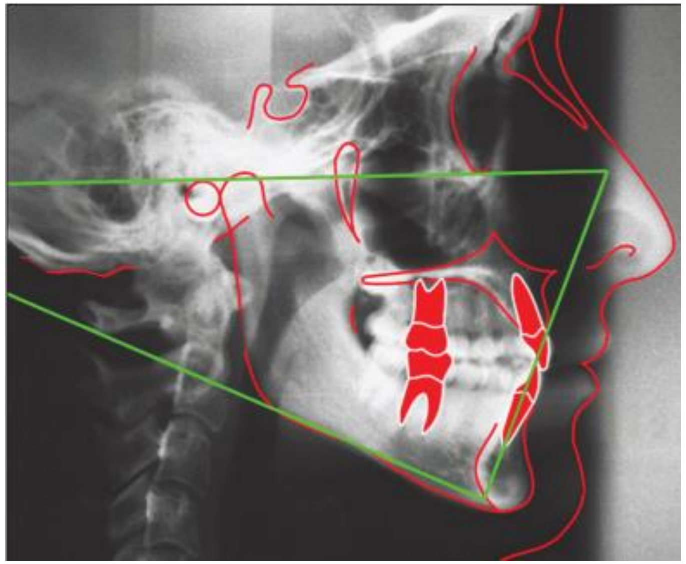




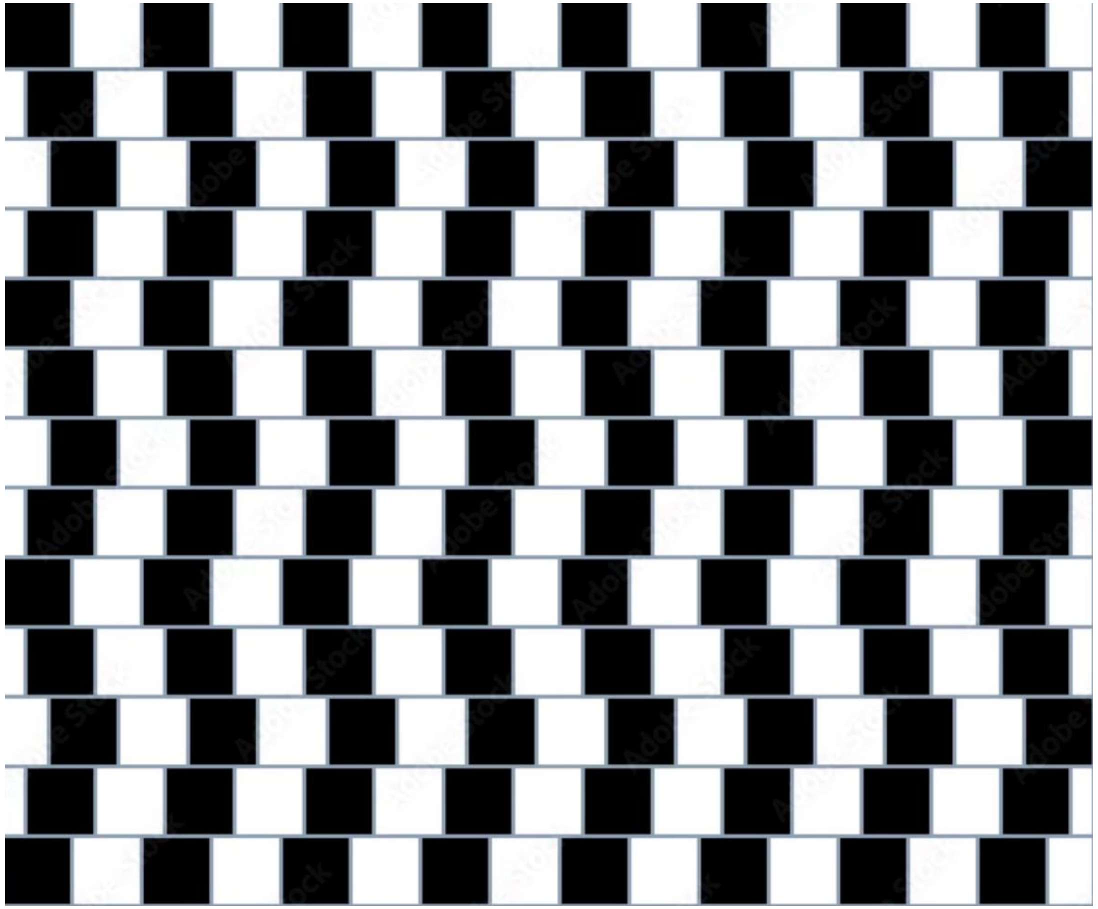
What is Cephalometry

- **Cephalometry** is a standardized radiographic diagnostic method used mostly in **orthodontics** to study the **relationships between the skull, jaws, teeth, and soft tissues** using a standardized **lateral skull X-ray**, called a **cephalometric radiograph**.





Why Do We Need Cephalometry?



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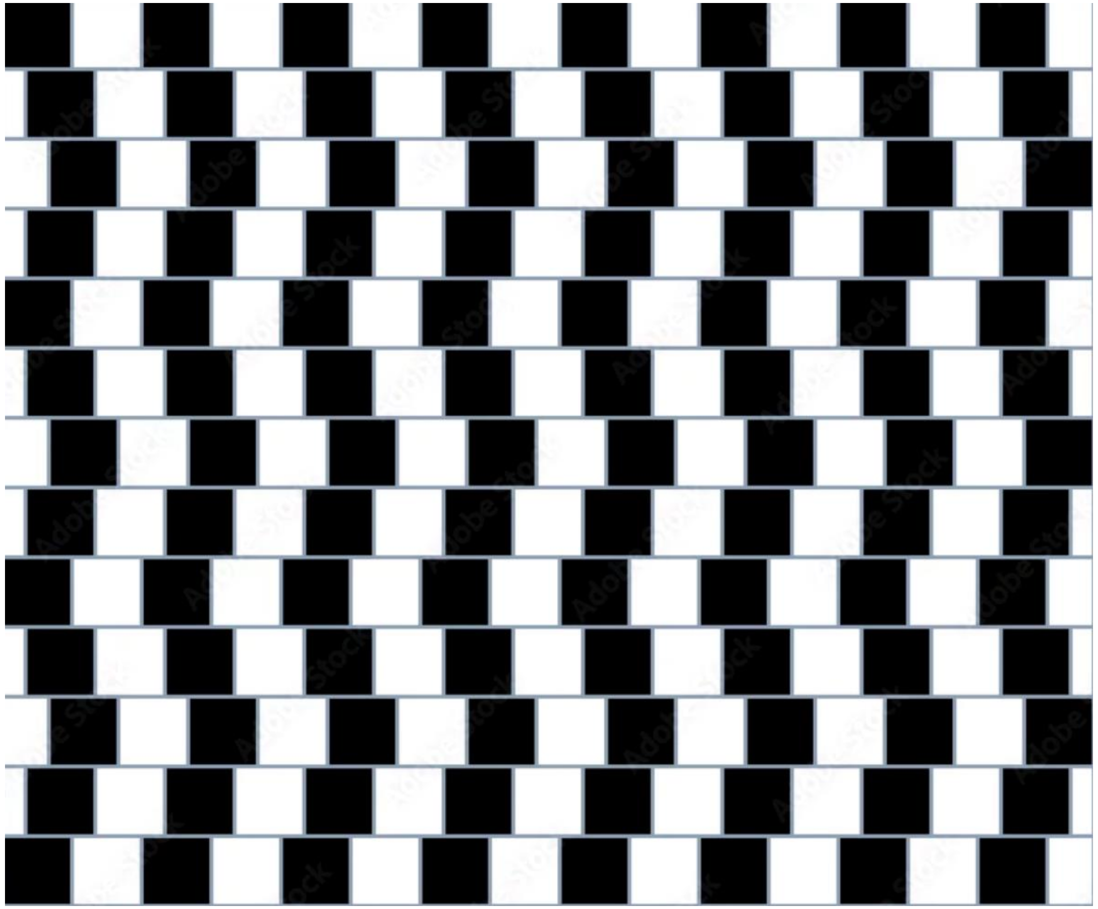

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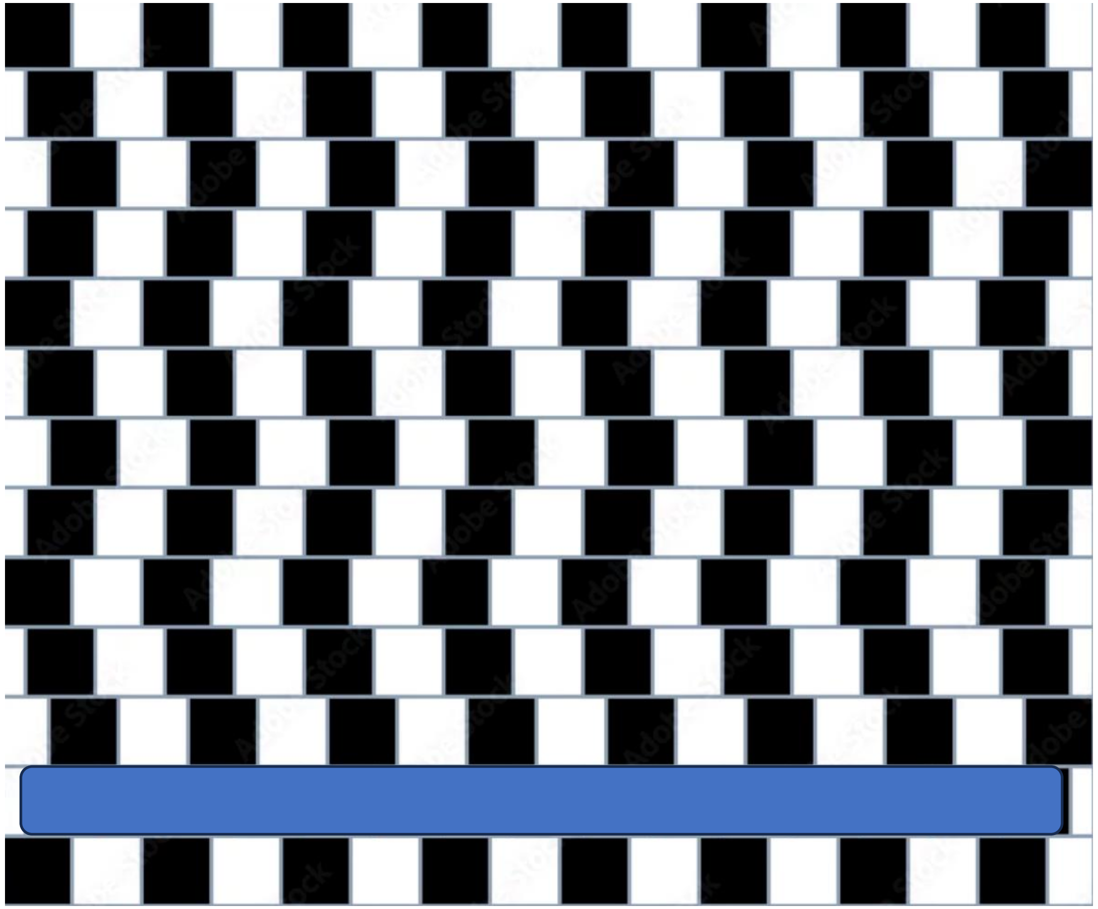
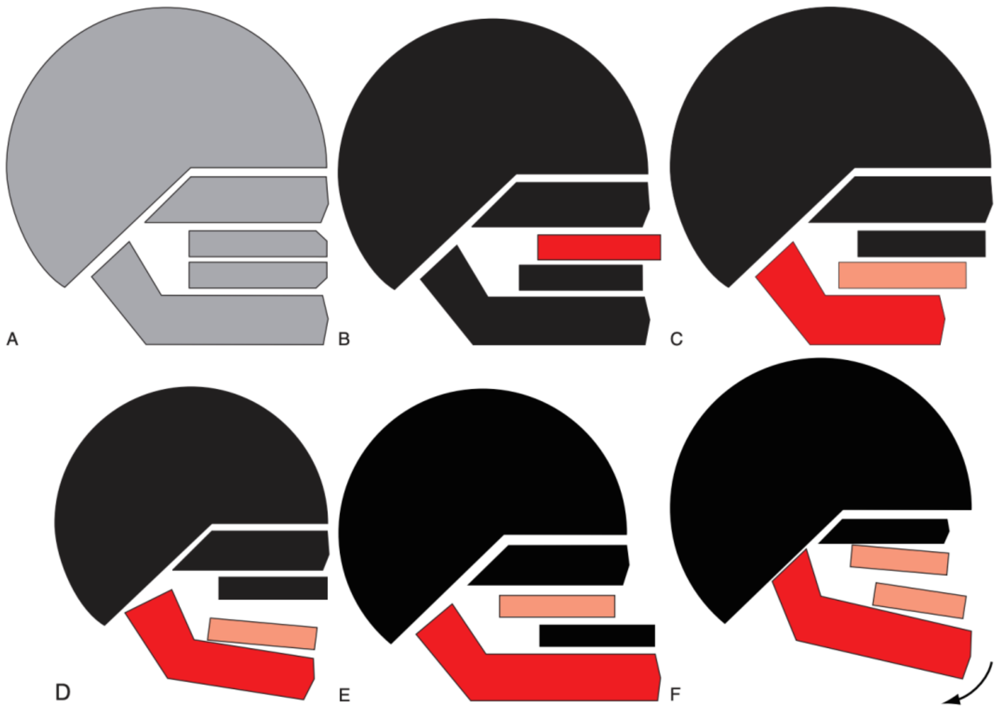


Why Do We Need Cephalometry?

- Before the advent of cephalometry in the early 1930s, orthodontic diagnosis relied largely on clinical examination and facial appearance. As a result, skeletal discrepancies were often oversimplified. Class III malocclusions were commonly attributed to mandibular prognathism, while Class II malocclusions were frequently assumed to result from maxillary overgrowth.

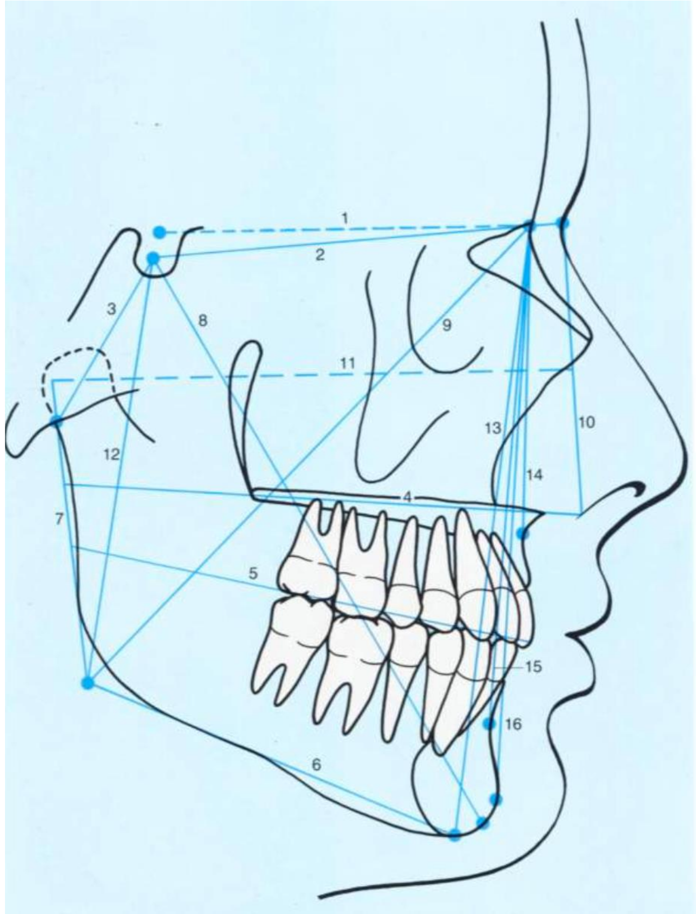
Why Do We Need Cephalometry?

- The introduction of cephalometric analysis transformed orthodontic diagnosis by allowing objective evaluation of maxillary and mandibular positions, growth patterns, and dentoalveolar compensations, leading to a more accurate understanding of the multifactorial nature of malocclusions.



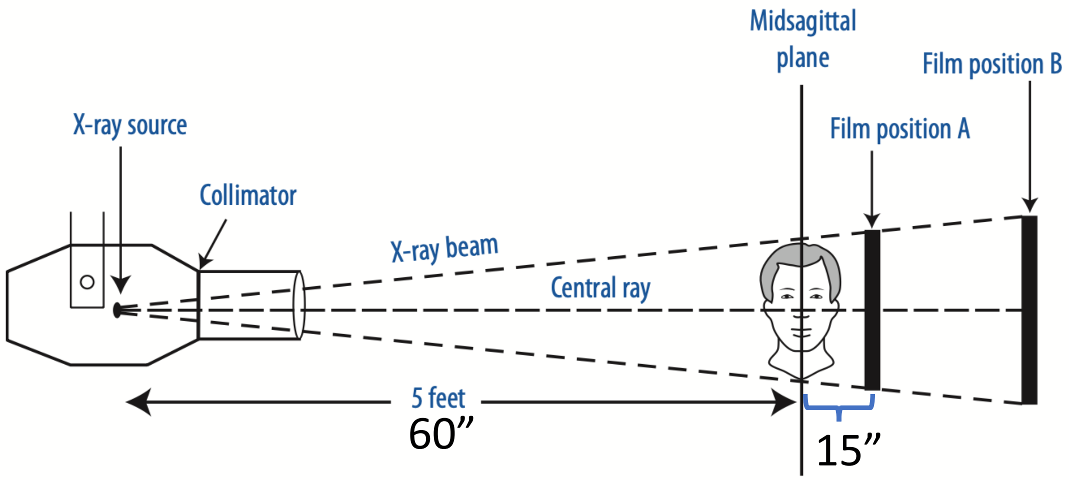
Cephalometry Allows Us to:

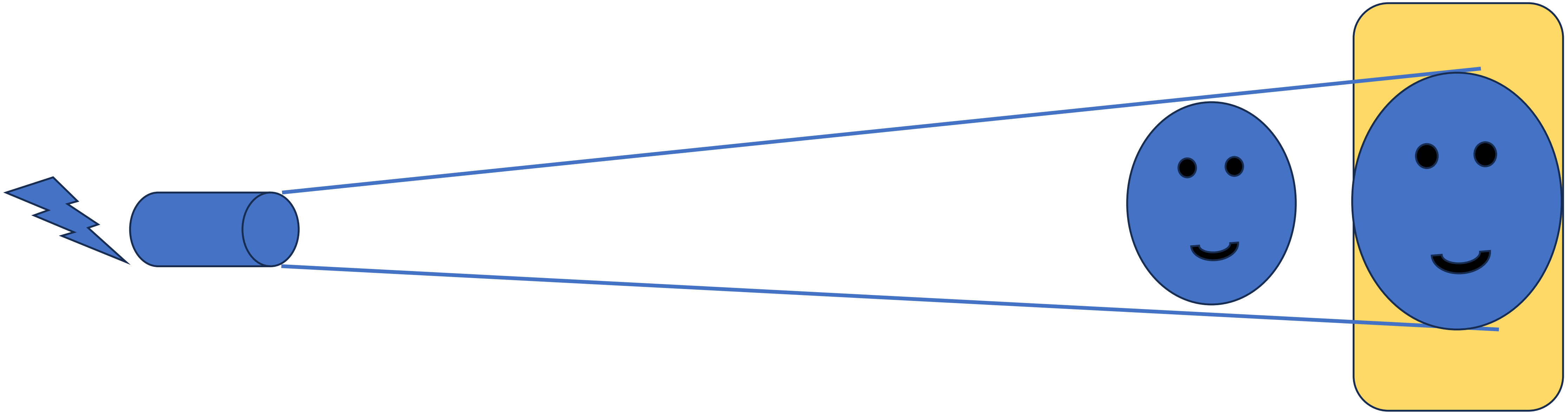
- Evaluate **jaw position** (maxilla and mandible)
- Determine Components of **Class I, II, or III Malocclusion**
- Assess **vertical growth pattern** (long face vs short face)
- Analyze **incisor position and inclination**
- Evaluate **facial profile and soft tissues**
- Monitor **growth and treatment changes over time**

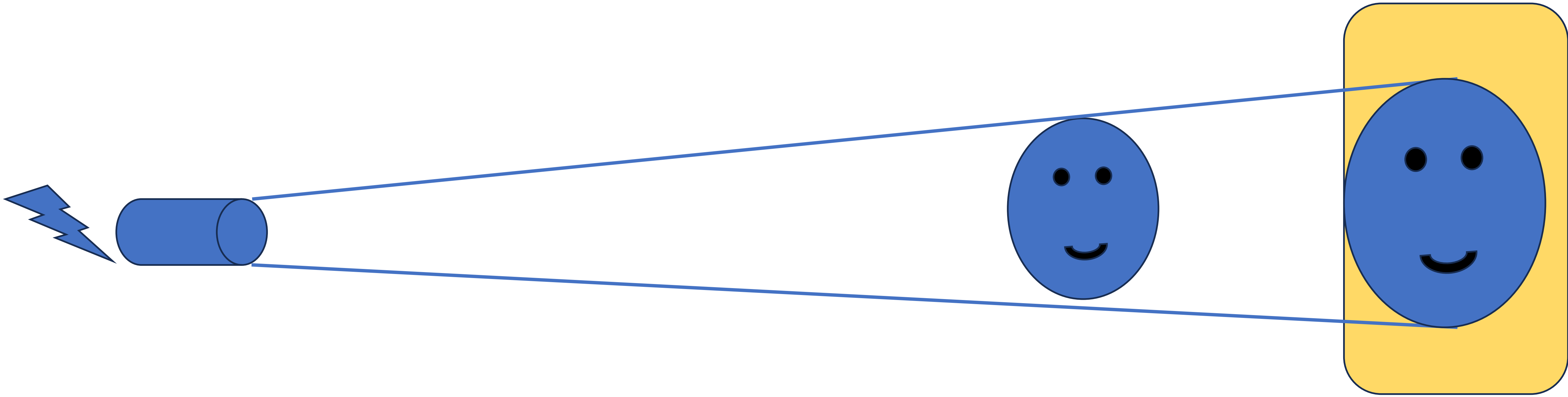


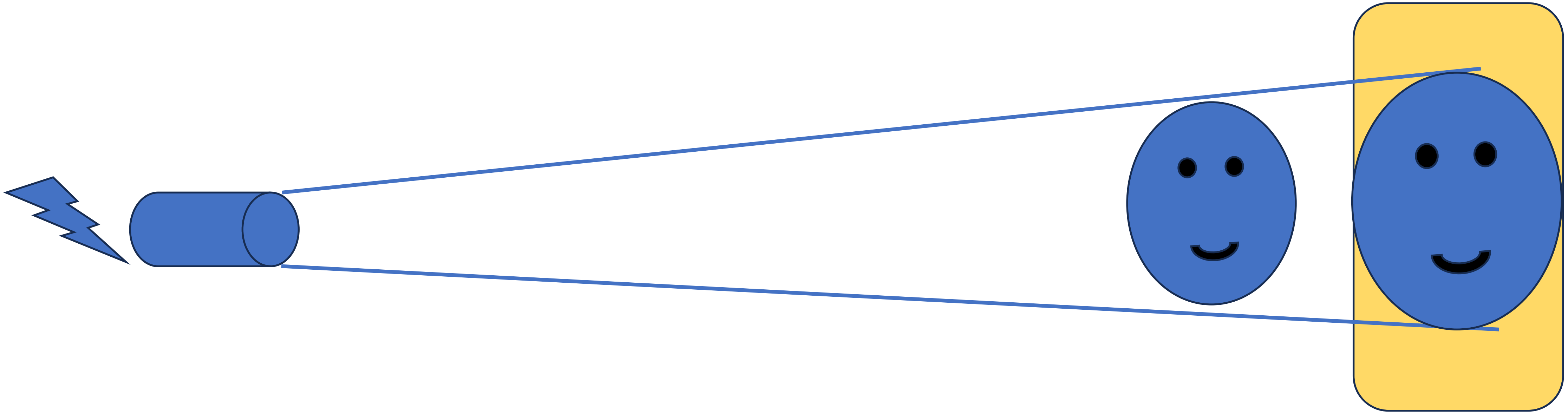
Cephalometry

- The patient's head is positioned in a **standardized way**
- The distance between the X-ray source and patient is fixed
- This allows **reproducible measurements**, so images can be compared over time









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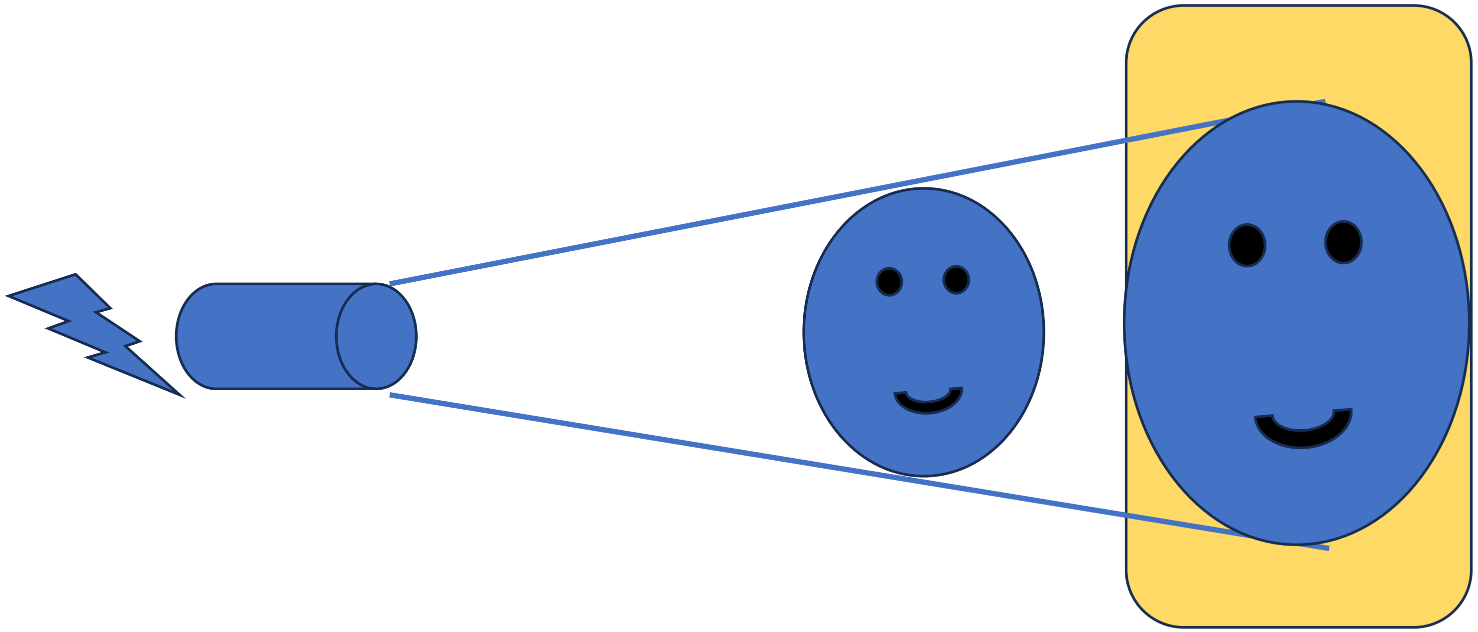
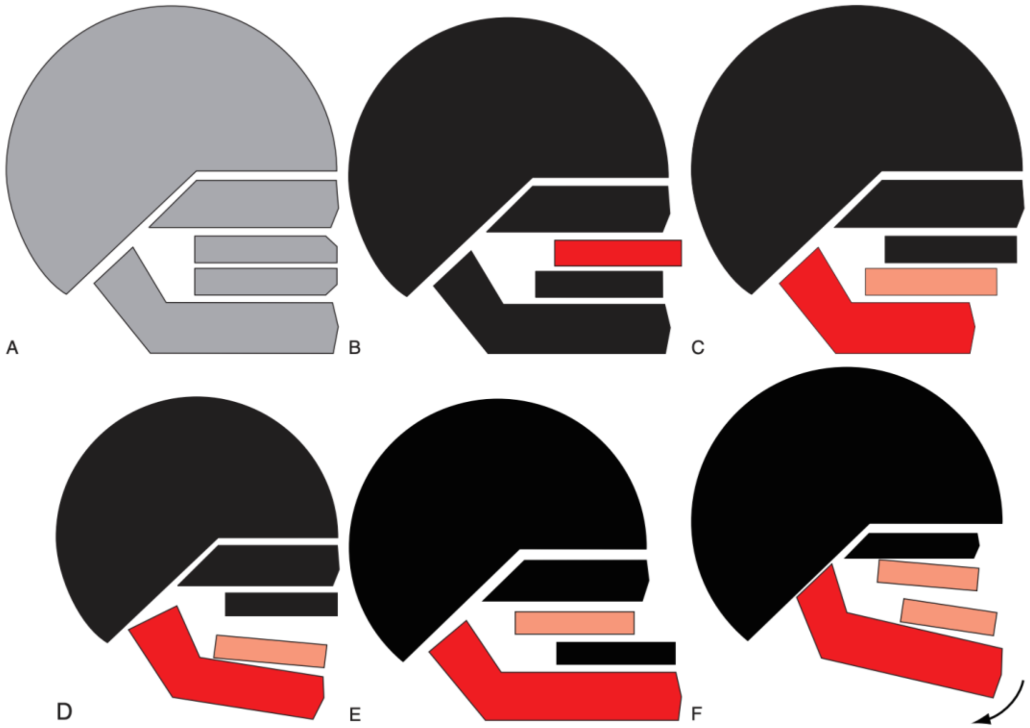


Cephalometric Analysis Components



1. Anterior Cranial base
2. Maxillary base
3. Maxillary dento-alveolar process
4. Mandibular base
5. Mandibular dento-alveolar process

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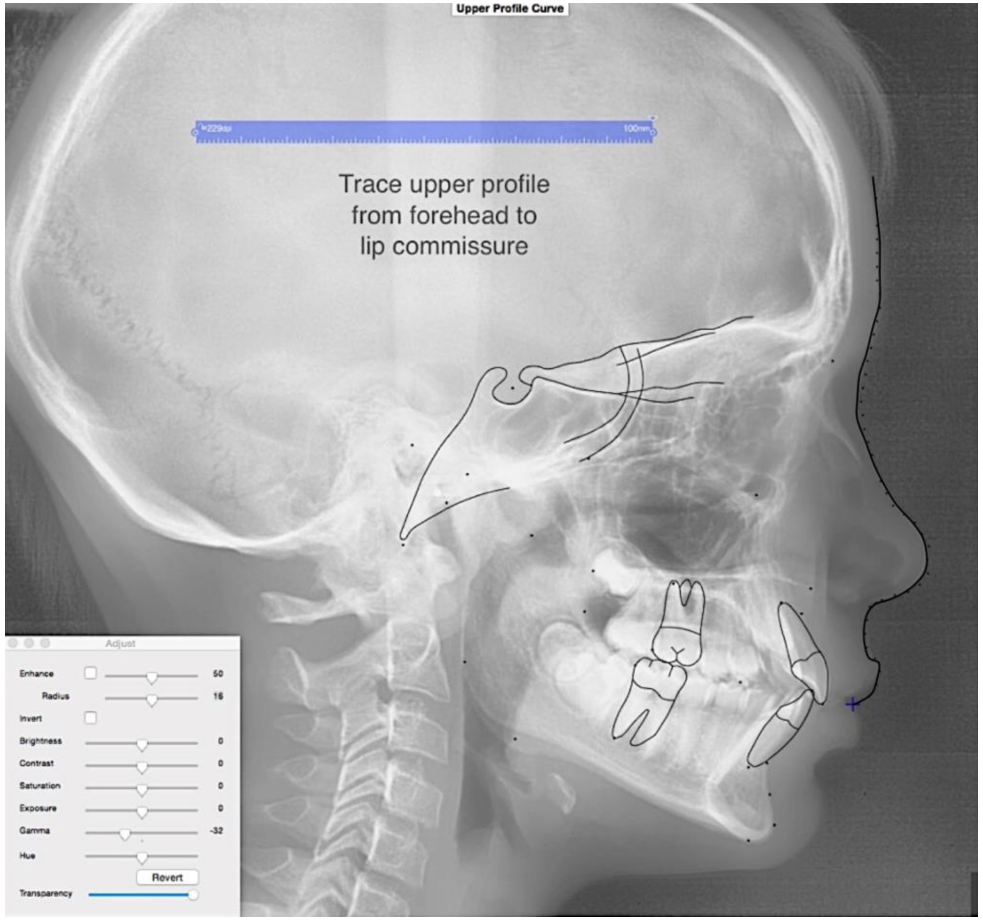
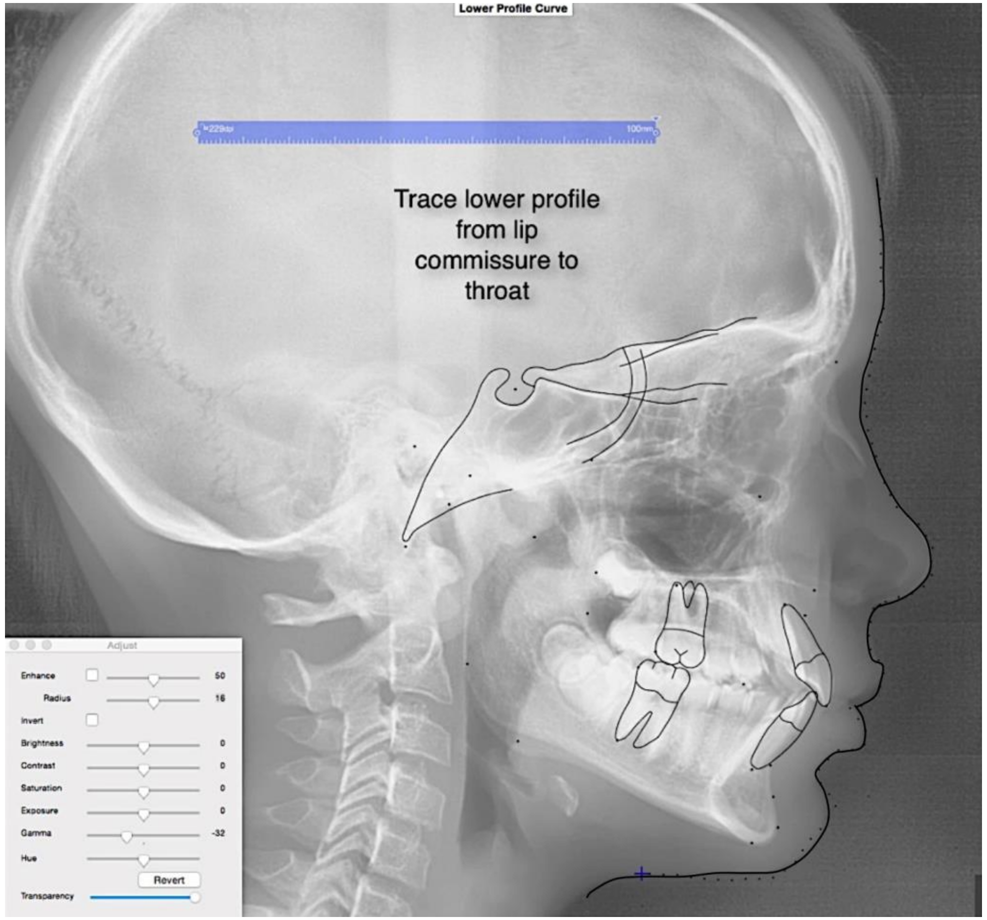
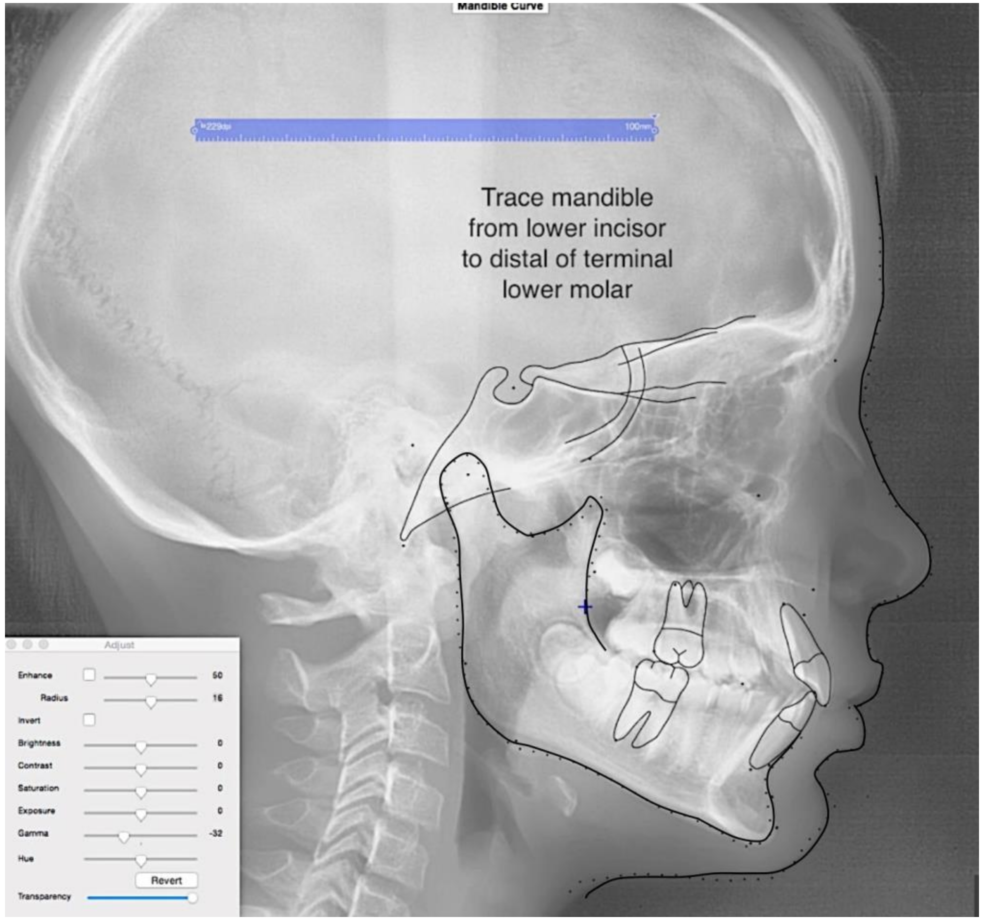
Cephalometric Tracing

- Tracing is the process of creating an overlay drawing from a cephalometric radiograph to identify anatomical landmarks and construct reference lines for analysis. It translates complex radiographic shadows into a simplified geometric map used for measurements



Cephalometric Tracing





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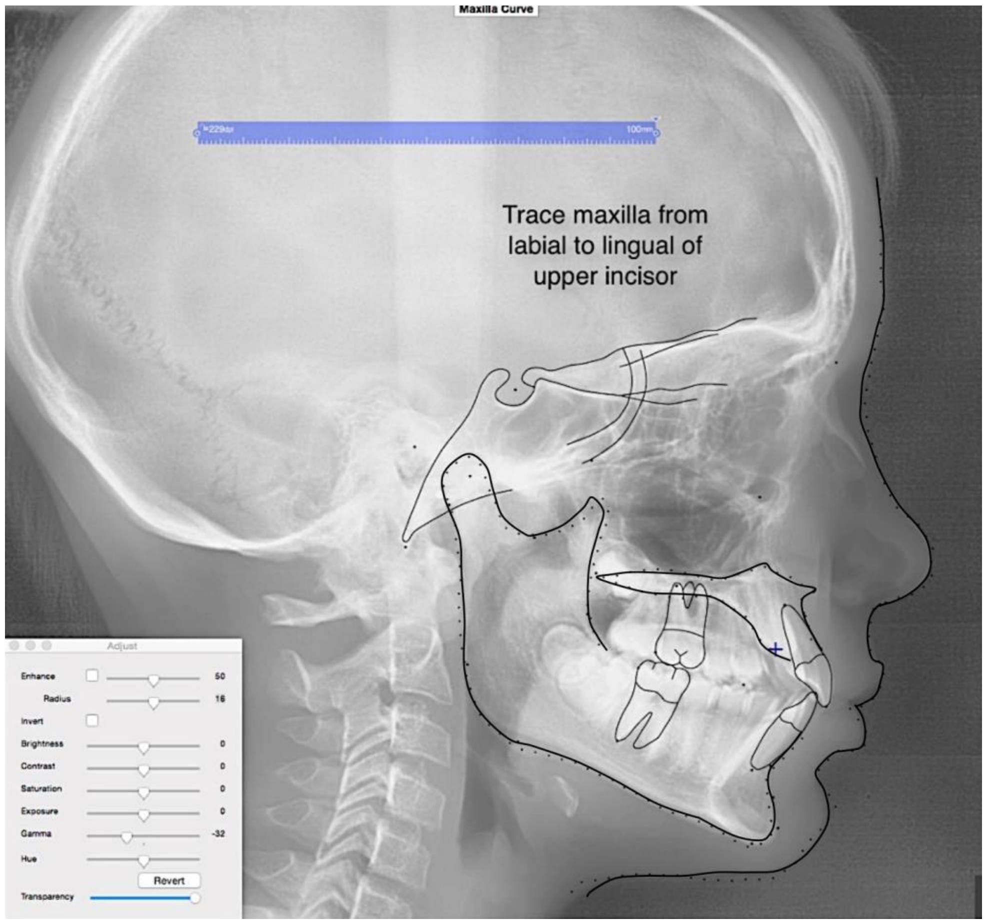

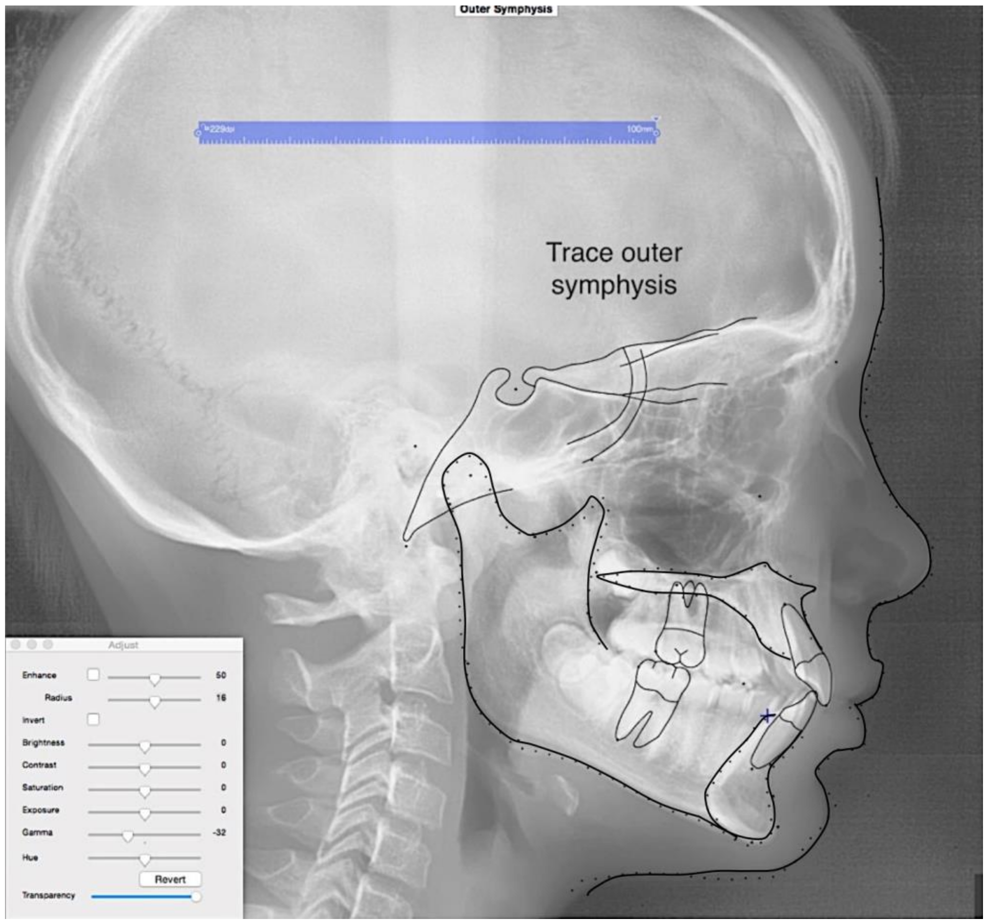

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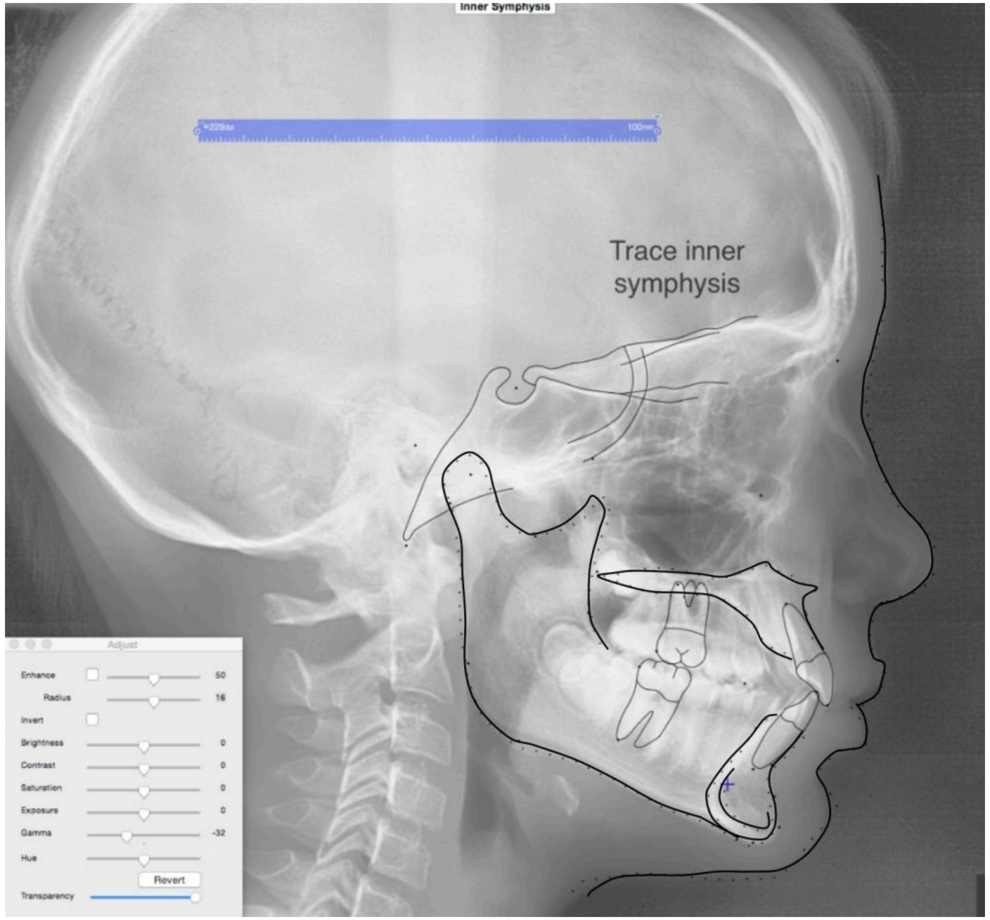

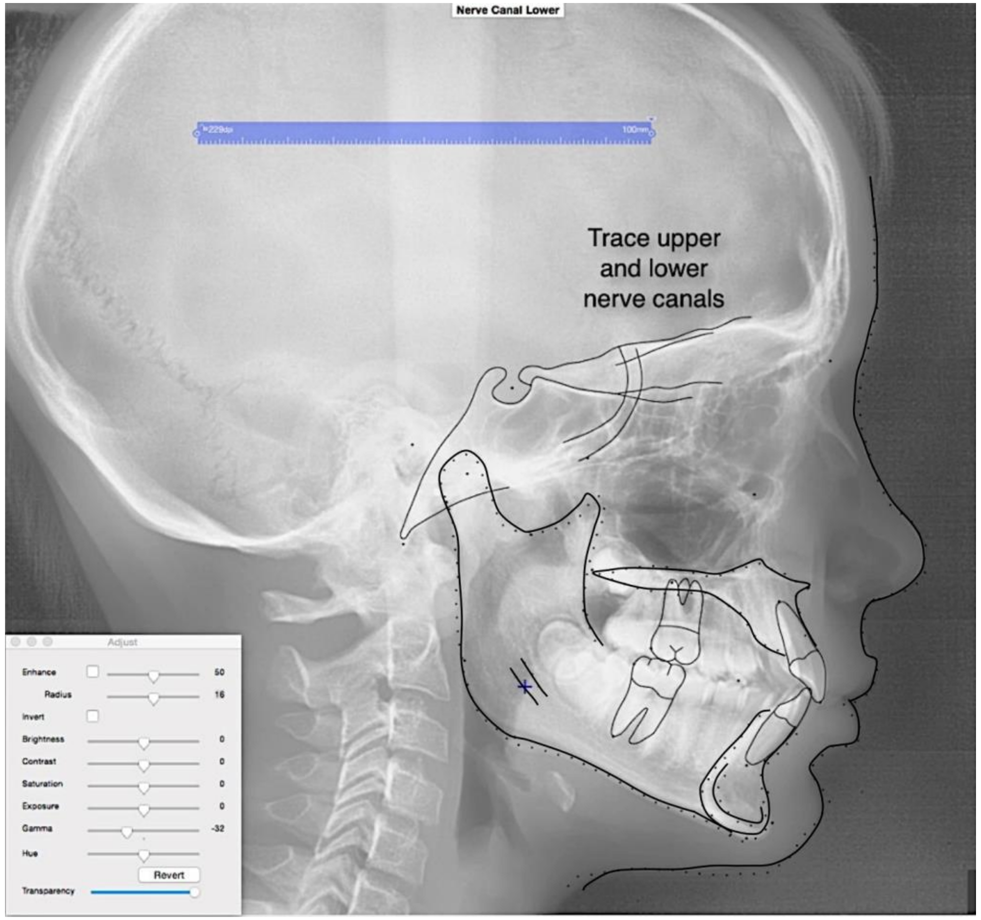

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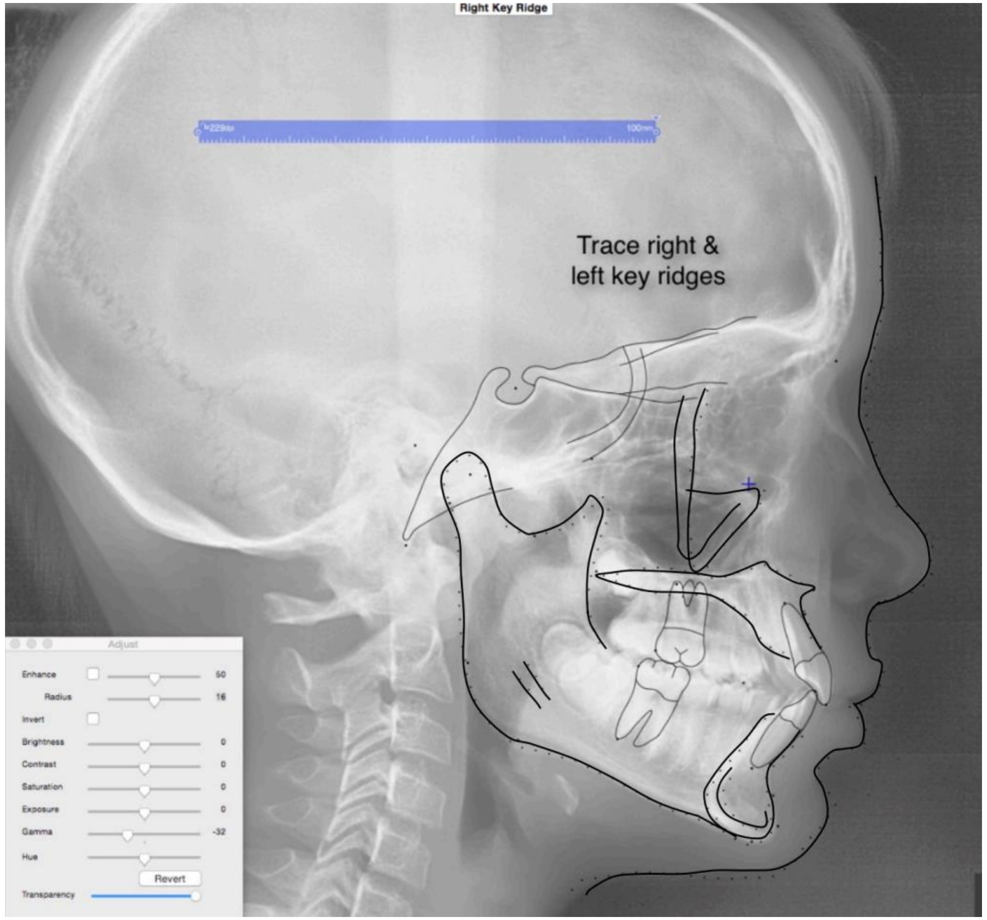

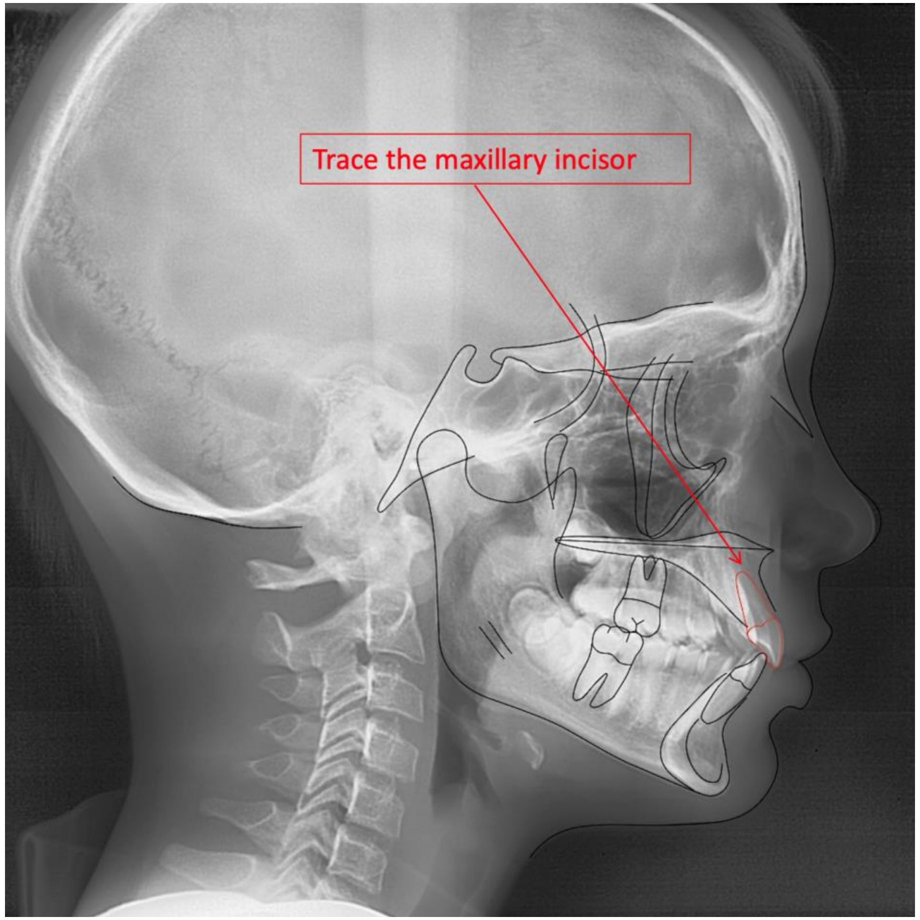

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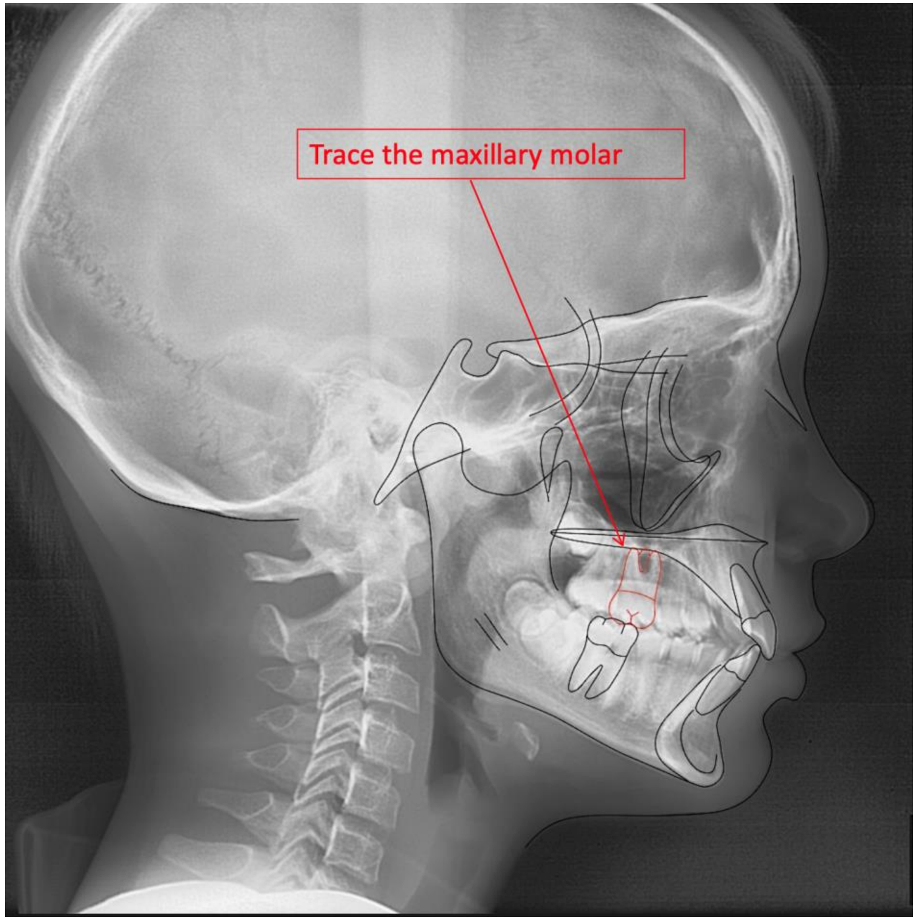

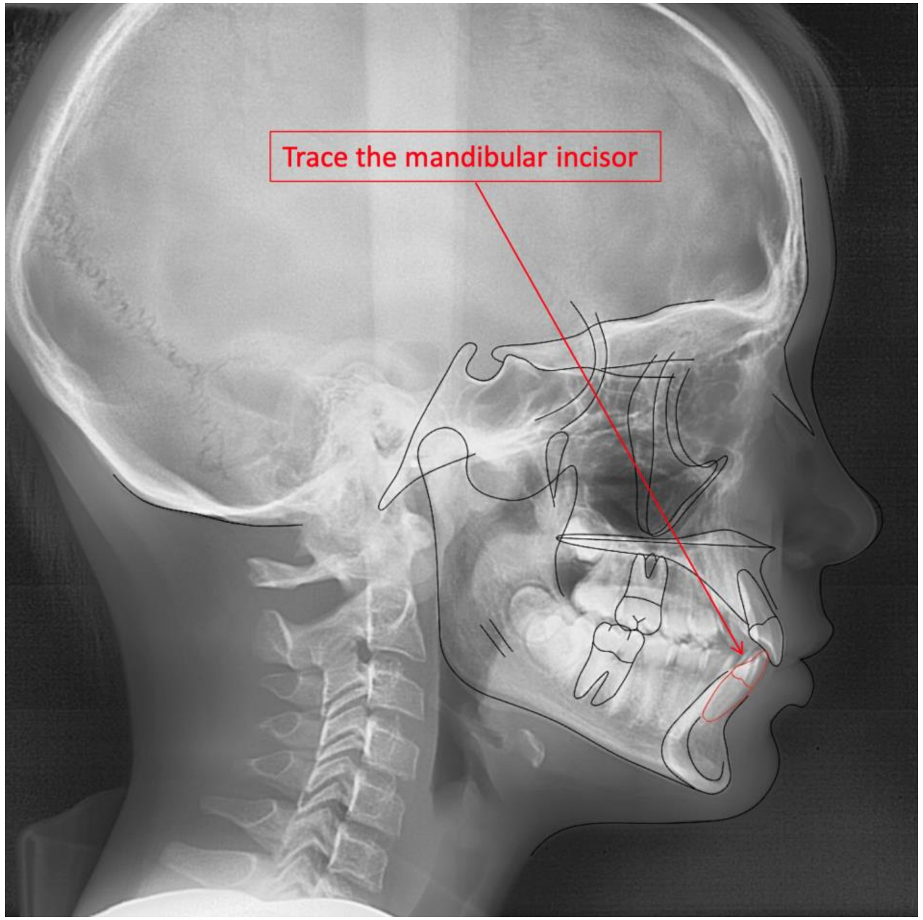

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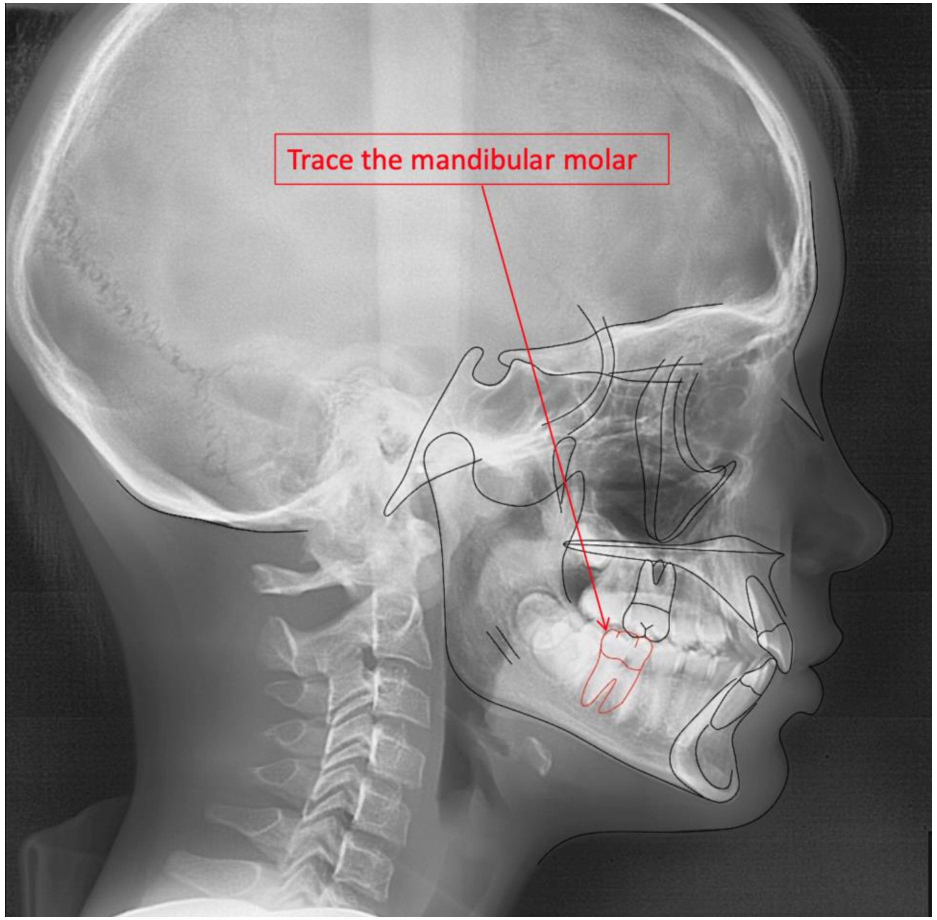

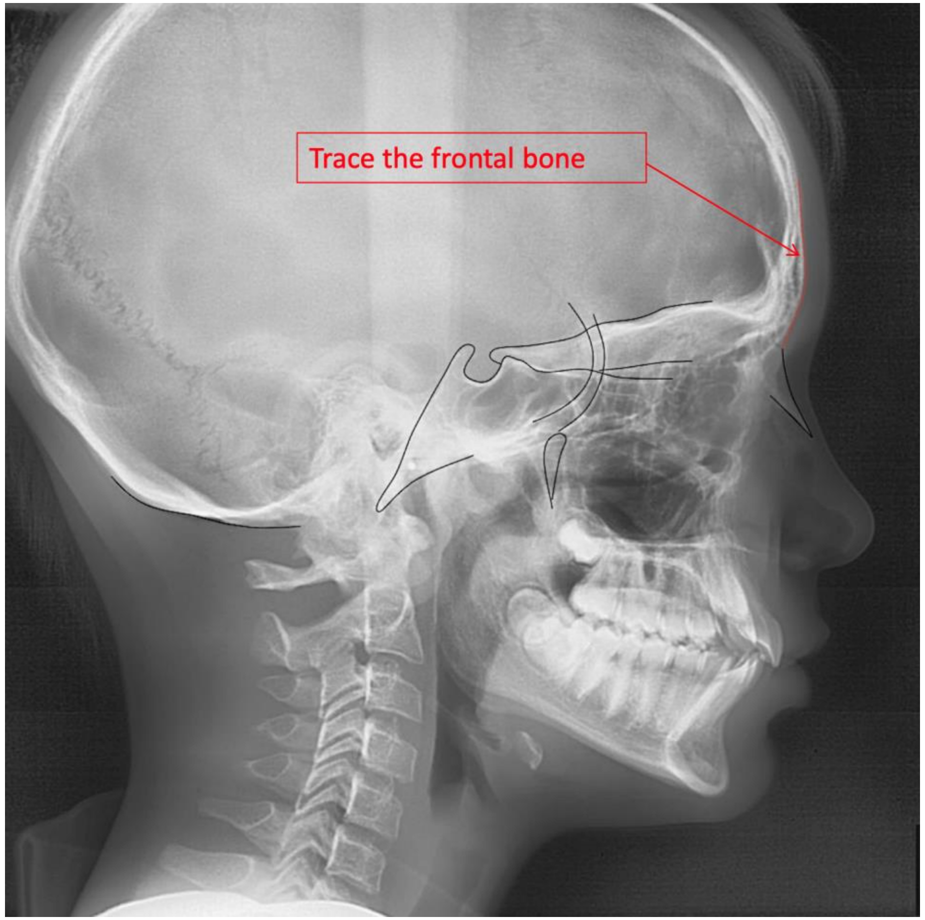

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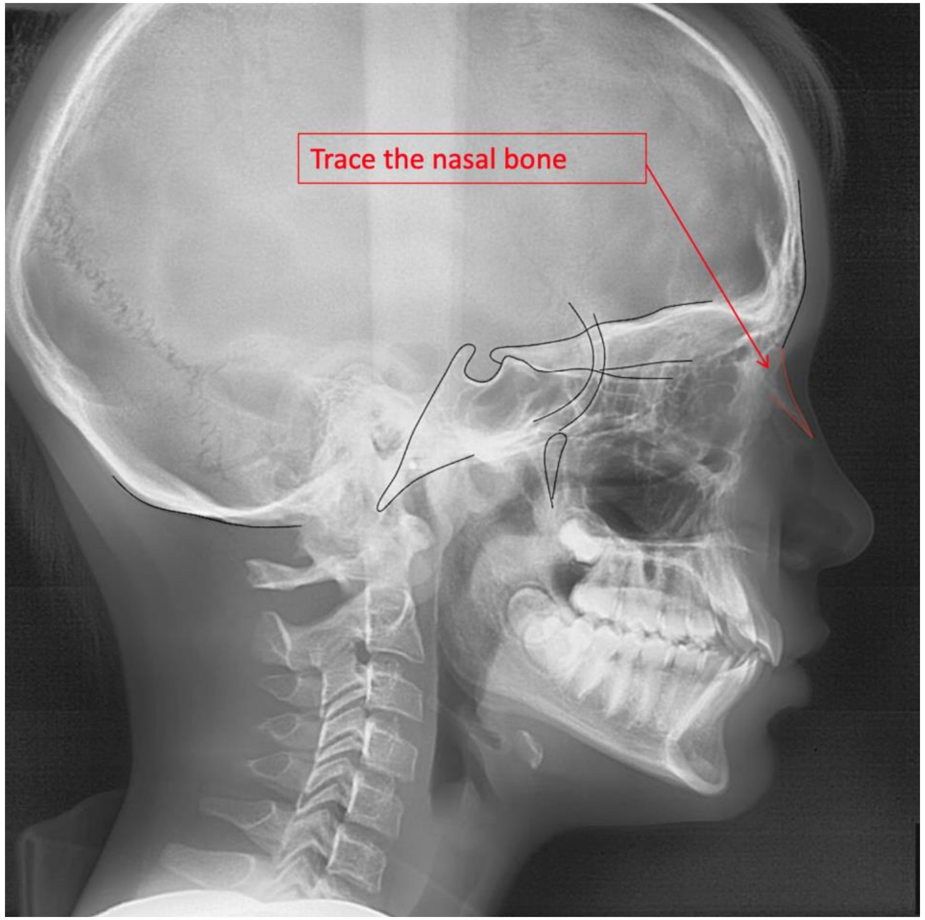
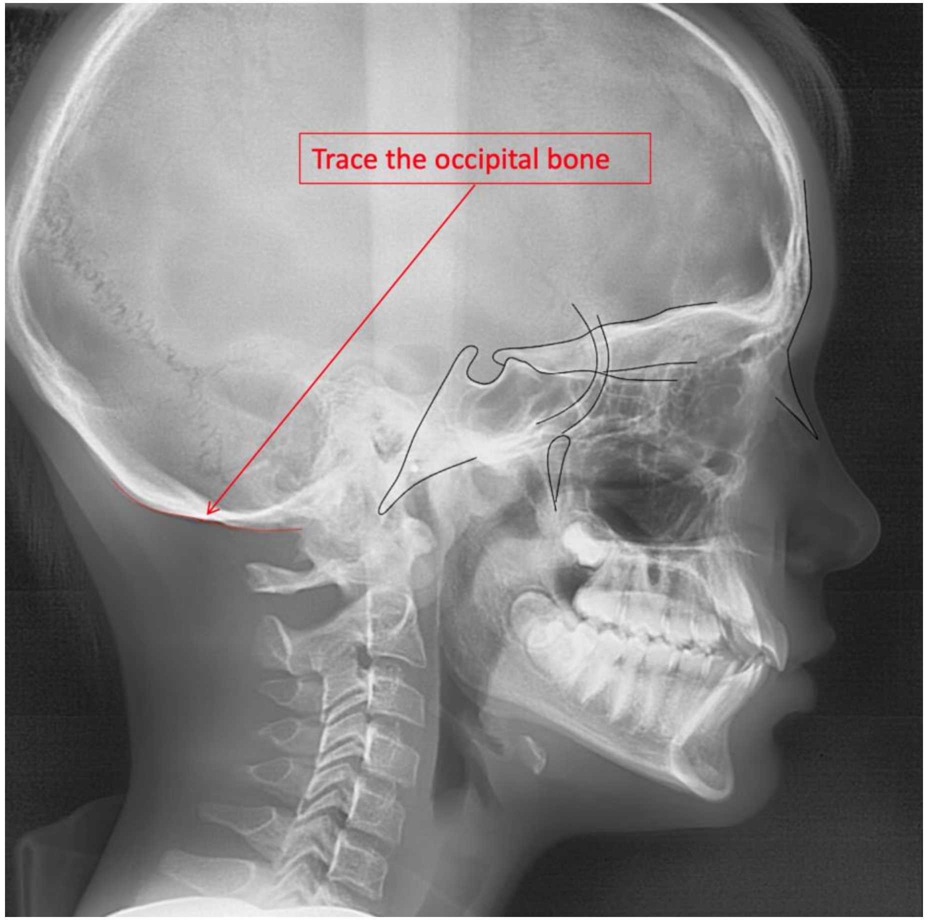
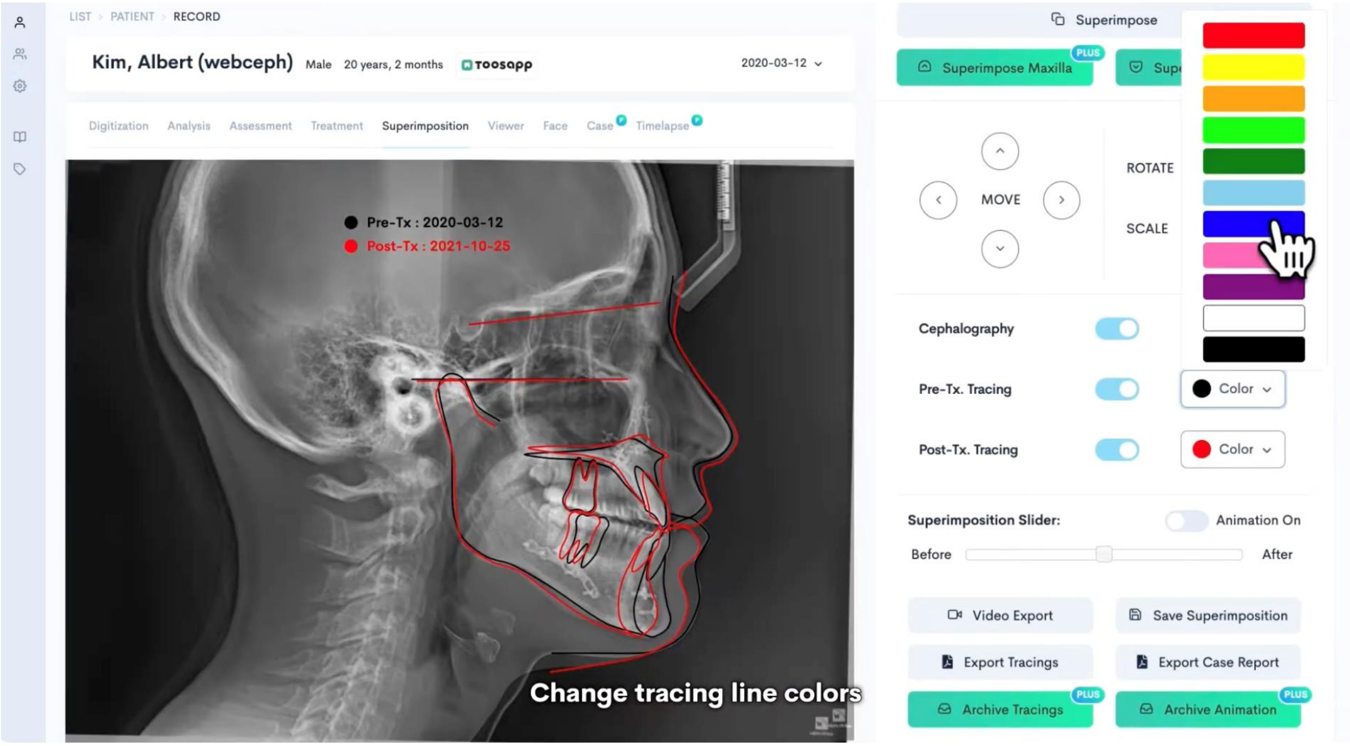




Cephalometric Tracing

- Original tracing is black
- Progress tracing is blue
- Final tracing is red



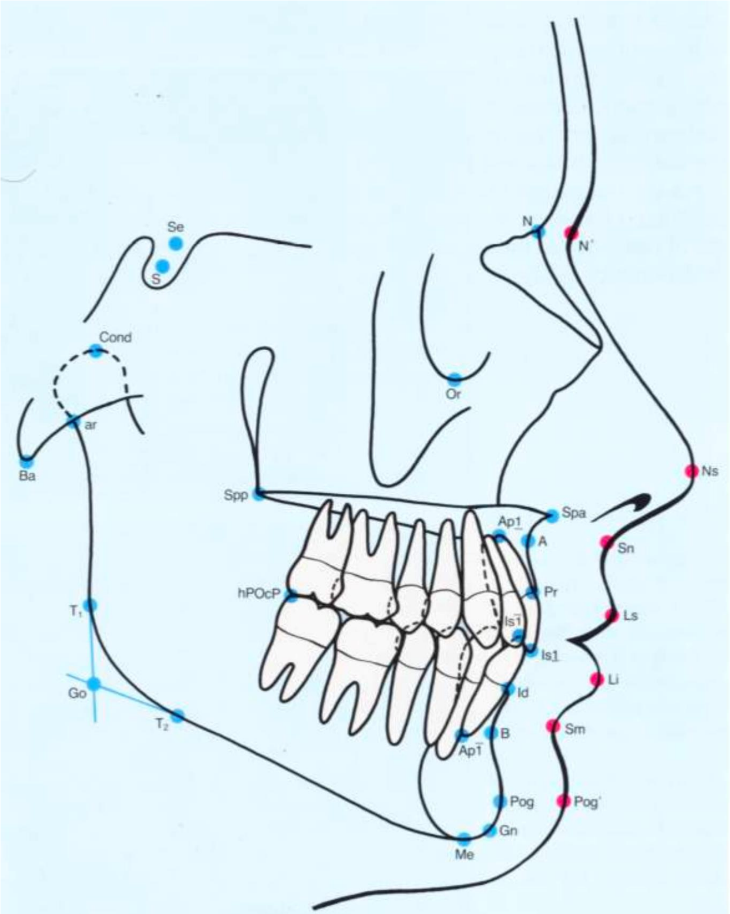
Cephalometric Digitization

- The process of identifying and marking specific anatomical landmarks on a digital radiograph displayed on a computer screen.
- Digitization is the recording of the **X and Y coordinates** of cephalometric landmarks into a computer system.

Cephalometric Landmarks

1. **<u>Anatomic:</u>** Points that correspond to actual physical anatomical structures. Example: Anterior Nasal Spine (ANS)

2. **<u>Constructed:</u>** Points obtained secondarily by extending lines or intersecting anatomical planes. Example: Articulare (Ar)



N = Nasion; the most anterior point of the frontonasal suture in the mid-sagittal plane

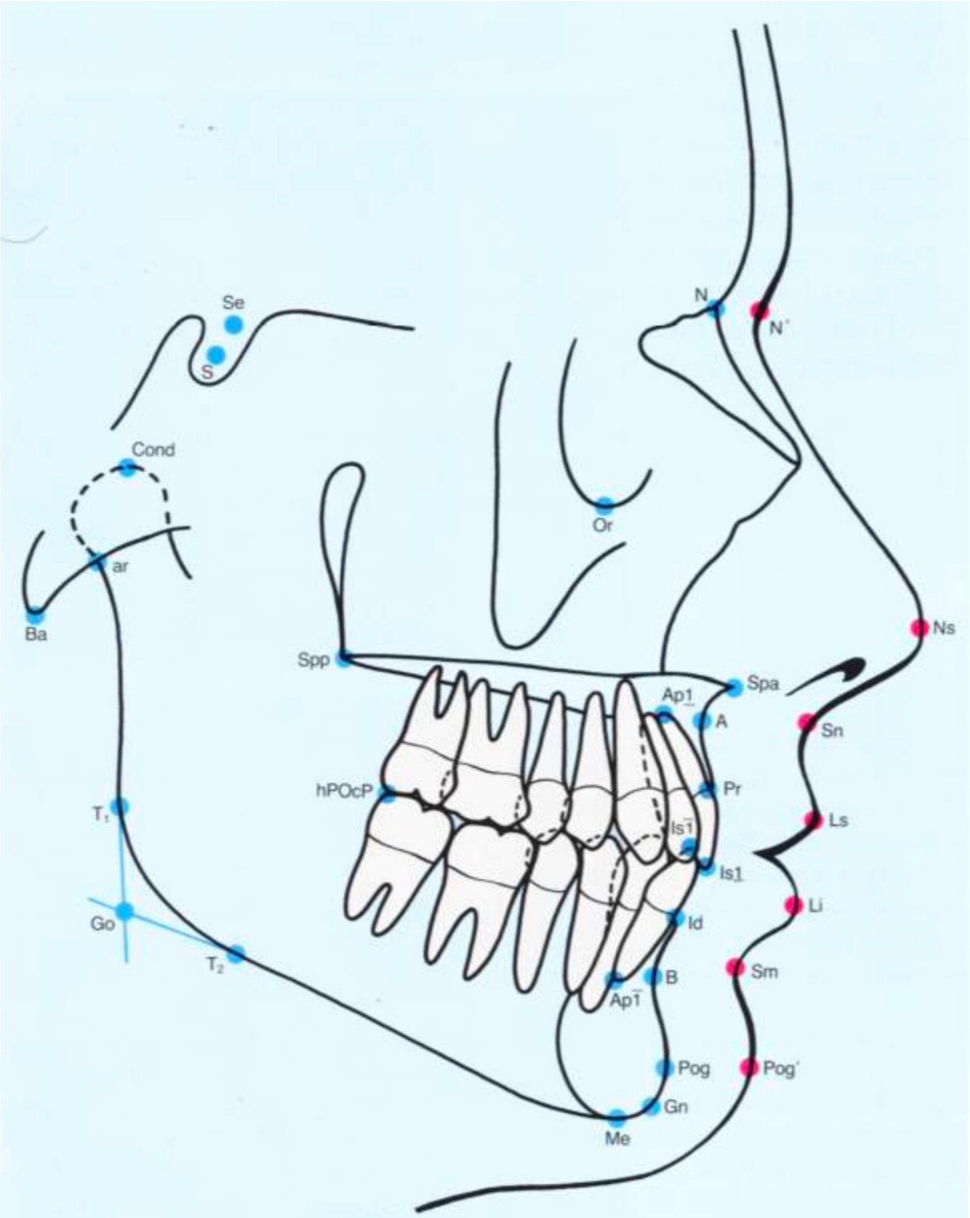
S = Midpoint of sella turcica

Se = Midpoint of the entrance to the sella

Cond = Condylion; the most posterior superior point of the condyle

Ar = Articulare; a constructed point at the intersection of the images of the posterior margin of the ramus and the outer margin of the cranial base

Ba = Basion; lowest point on the anterior margin of the foramen magnum in the median plane



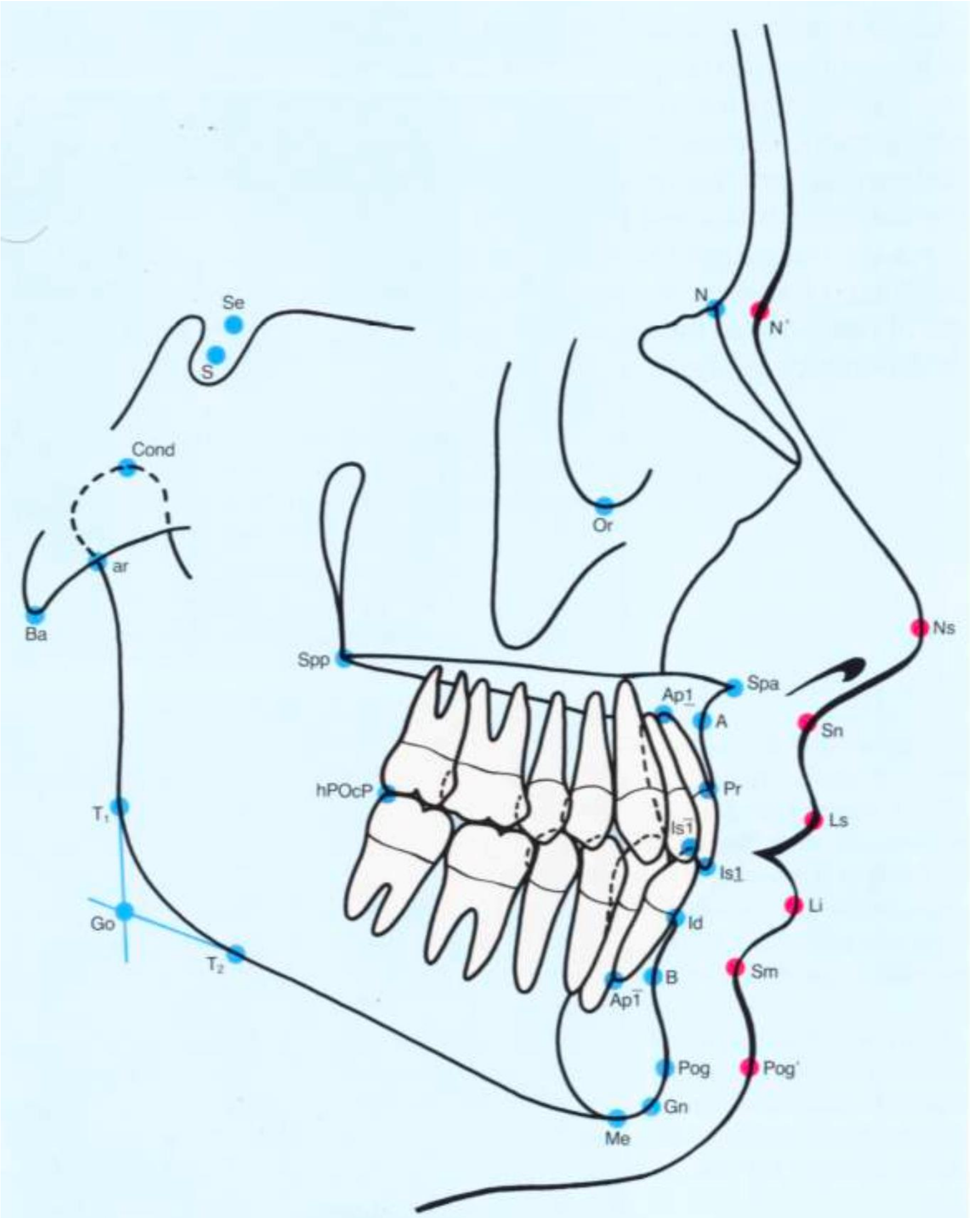
Go = Gonion; a constructed point at the intersection of the lines tangent to the posterior border of the ramus and the lower border of the mandible

Me = Menton; the most inferior point of the outline of the symphysis in the midsagittal plane

Gn = Gnathion; the most anterior inferior point on the bony chin

Pog = Pogonion; the most anterior point of the bony chin in the mid-sagittal plane

B = Point B, supramentale; the deepest point on the outer contour of the mandibular alveolar process between infradentale and pogonion



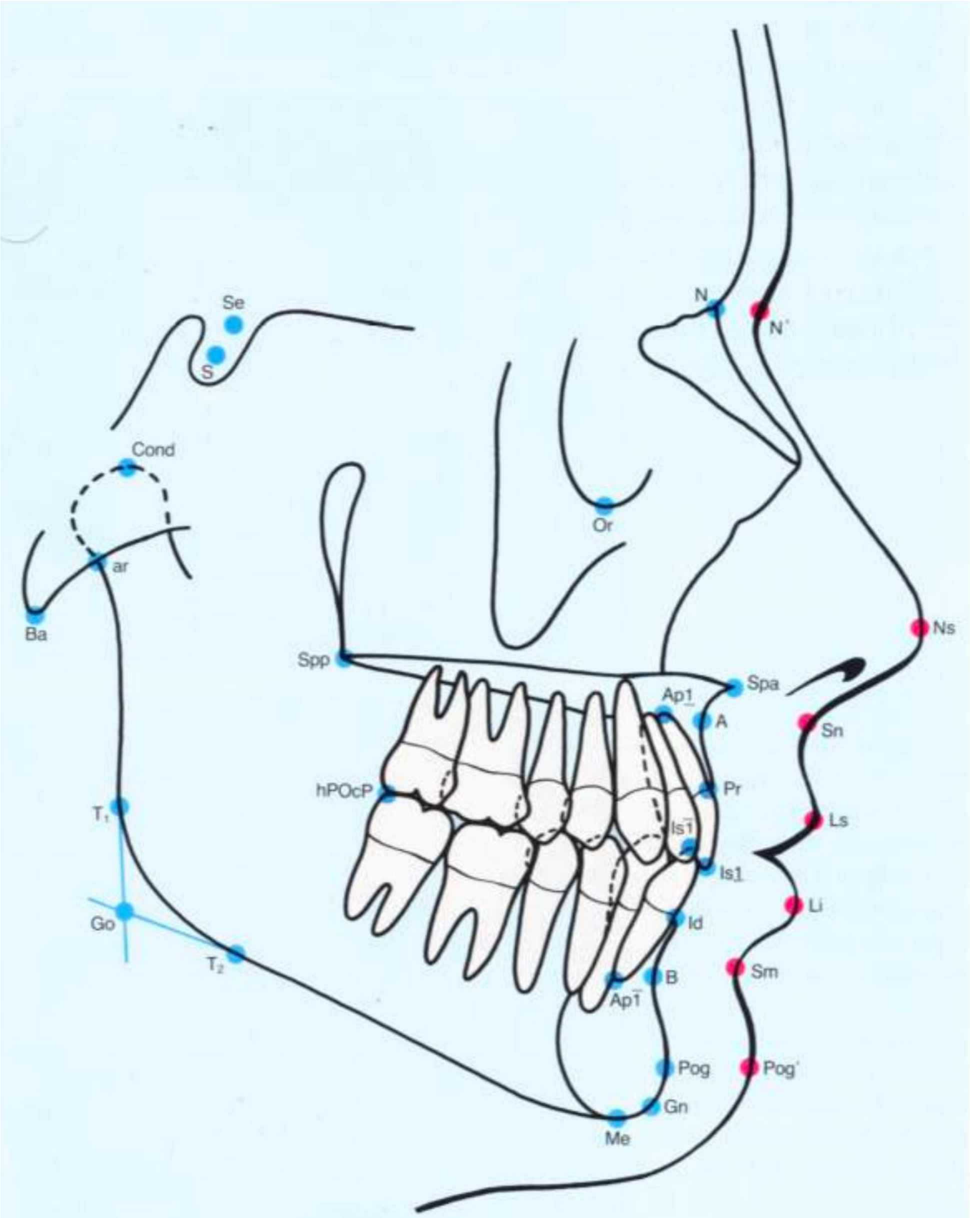
Id = Infradentale; the most anterior superior point on the alveolar process between the mandibular central incisors

Pr = Prosthion; the most anterior inferior point on the alveolar portion of the premaxilla between the upper central incisors

A = Point A, subspinale; the deepest midline point on the anterior outer contour of the maxillary alveolar process

ANS = Anterior nasal spine; the most anterior point of the tip of the anterior nasal spine in the midsagittal plane

PNS = Posterior nasal spine, The posterior spine of the palatine bone constituting the hard palate.



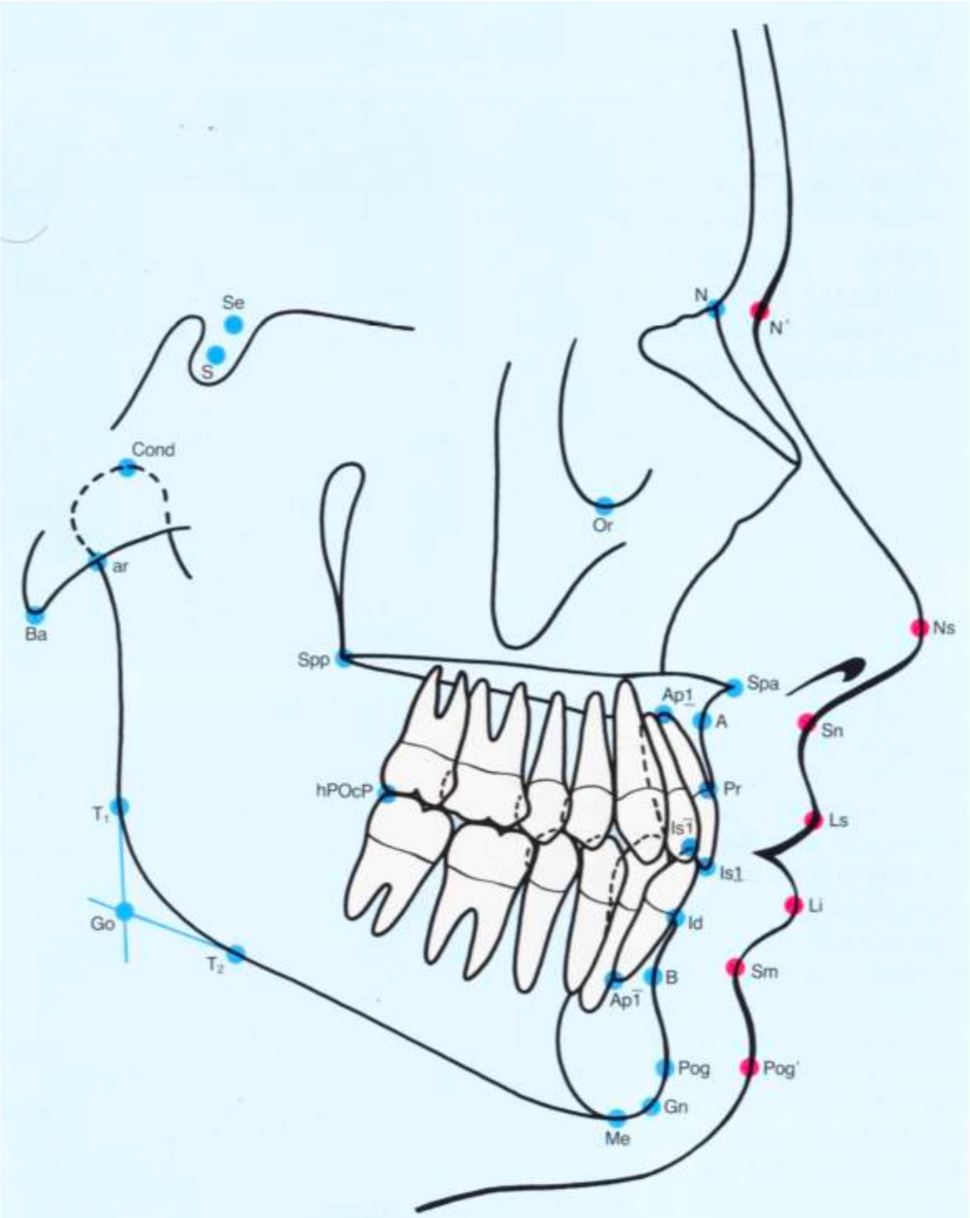
Or = Orbitale; lower-most point of the bony orbit

PPOcP = Posterior point of the occlusal plane; the most distal point of contact between the most posterior molars in occlusion

N' = Skin nasion; located at the point of maximum concavity between the nose and forehead

Ns = Tip of nose; the most anterior point of the soft-tissue nose

Sn = Subnasale; a skin point; the point at which the columella merges with the integument of the upper lip



Ls = Labrale superius; edge of upper lip

Li = Labrale inferius; edge of lower lip

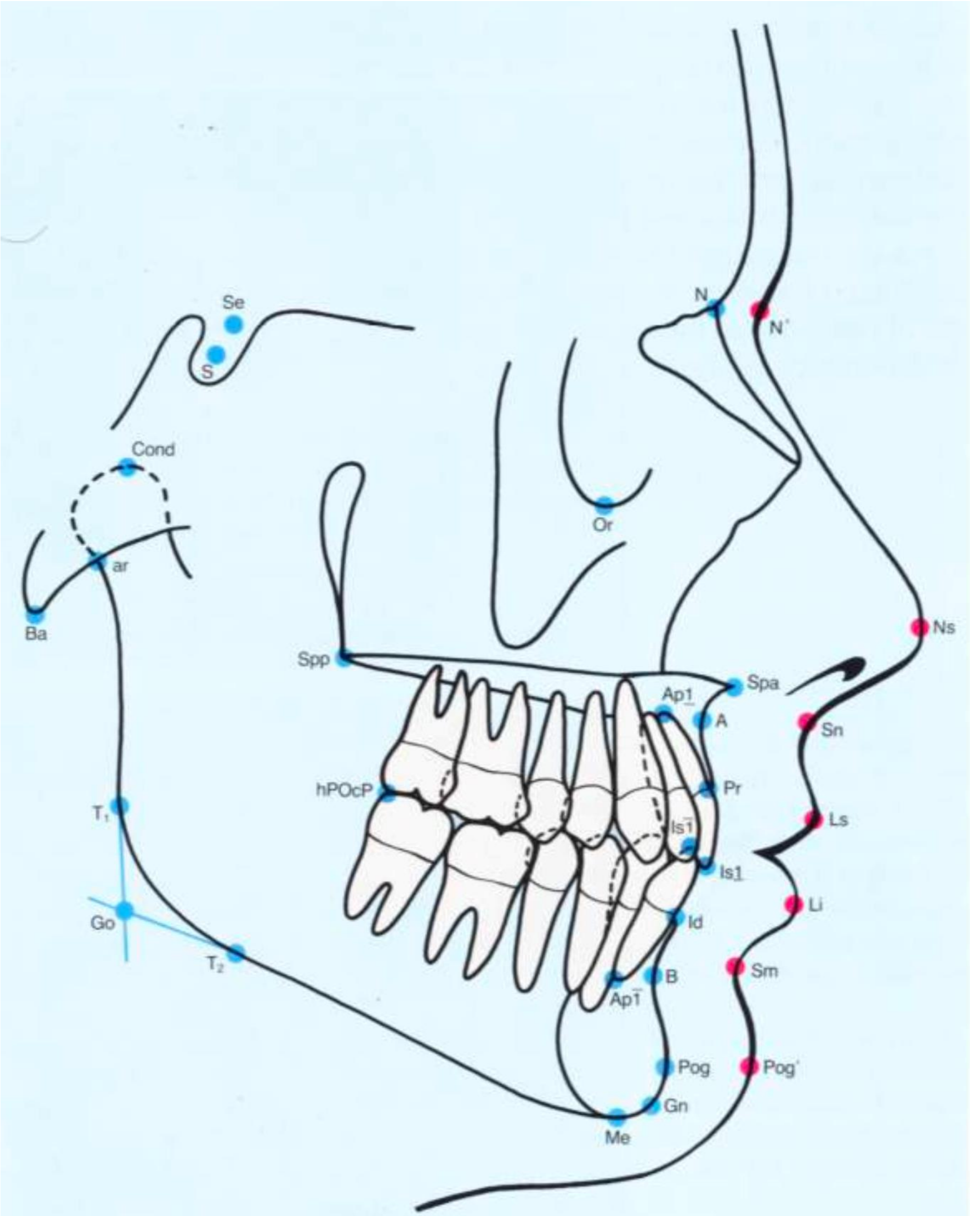
Sm = Supra mental, deepest

Point at which the labrale inferius

meets soft tissue pogonion

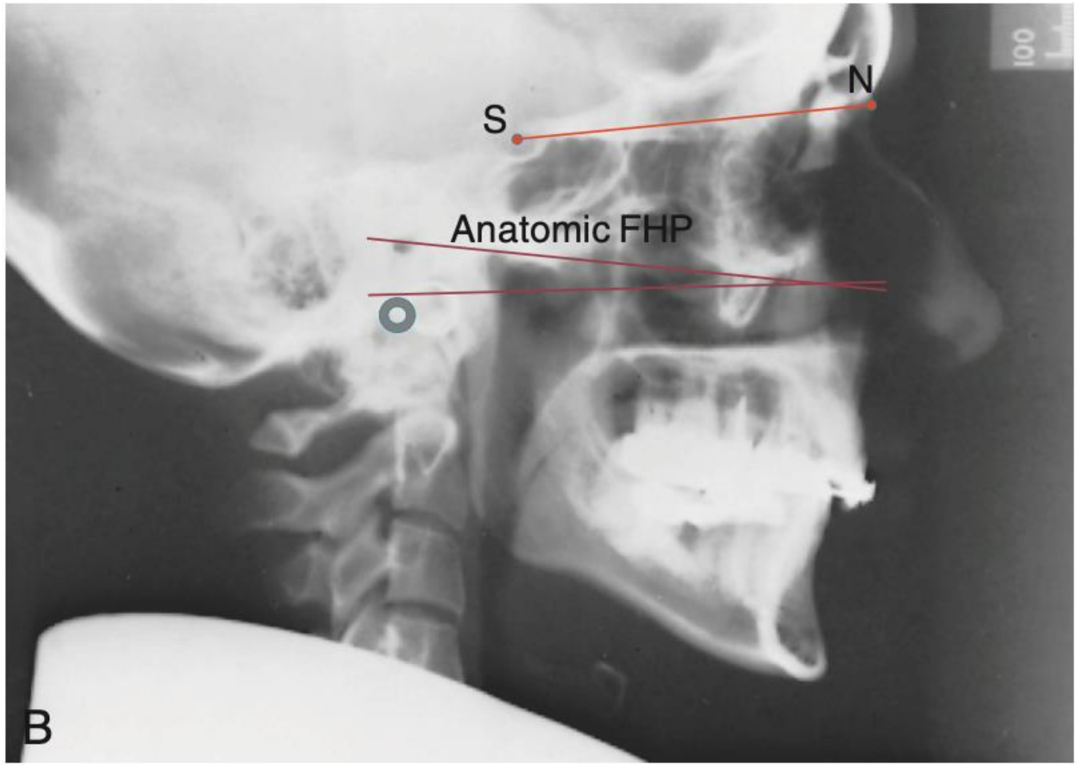
Pog' = Soft-tissue pogonion; the most

anterior point of the soft-tissue chin



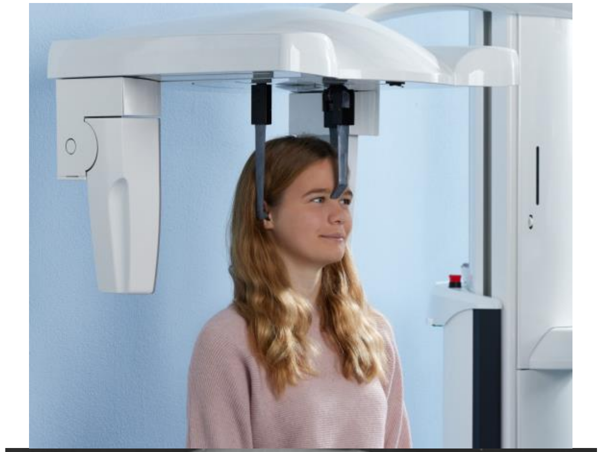
Cephalometric Reference Lines

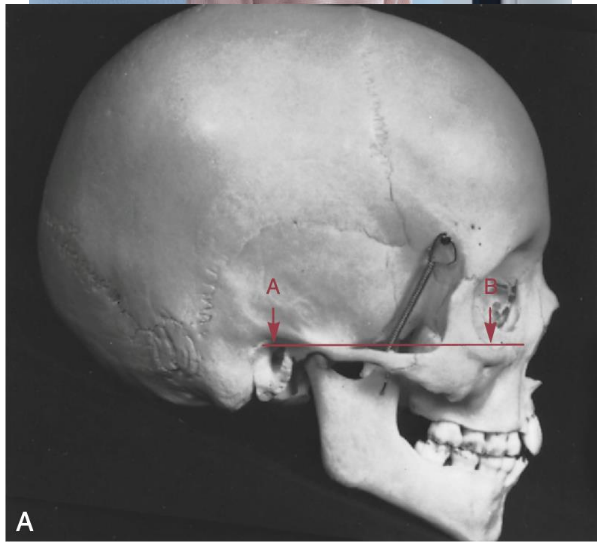
- In any technique for cephalometric analysis, it is necessary to establish a reference area or reference line.
  - SN
  - Frankfort



Cephalometric Reference Lines

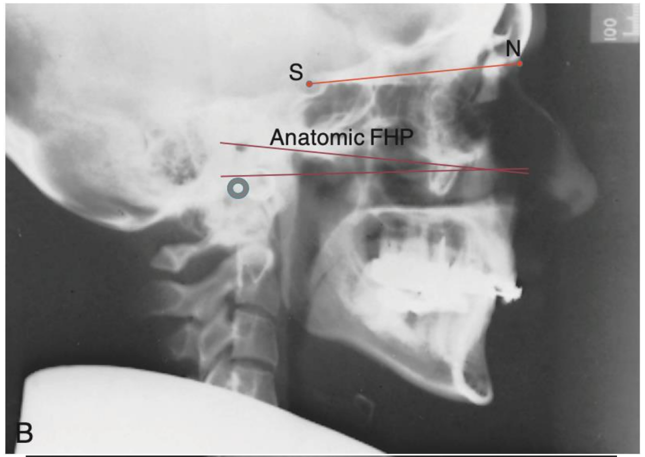
- Frankfort plane has two difficulties. The first is that both its anterior and posterior landmarks, particularly porion, can be difficult to locate reliably on a cephalometric radiograph. A radiopaque marker is placed on the rod that extends into the external auditory meatus as part of the cephalometric head positioning device, and the location of this marker, referred to as “machine porion,” is often used to locate porion.



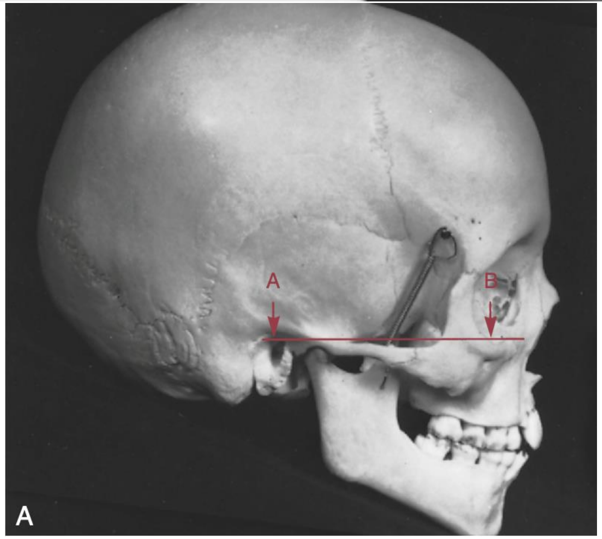


Cephalometric Reference Lines

- The shadow of the auditory canal can be seen on cephalometric radiographs, usually located slightly above and posterior to machine porion. The upper edge of this canal can also be used to establish "anatomic porion," which gives a slightly different (occasionally, quite different) Frankfort plane



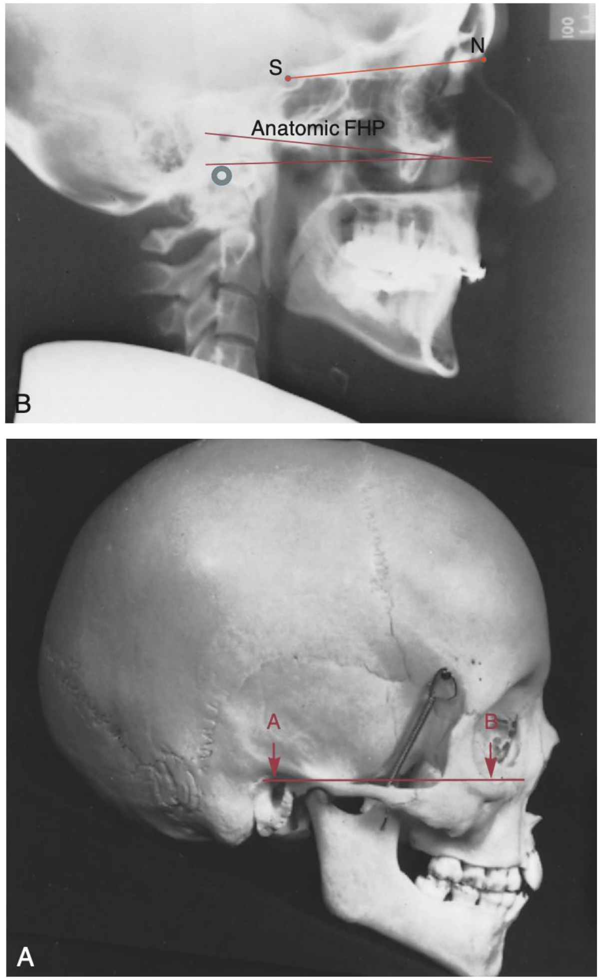
B



A

Cephalometric Reference Lines

- An alternative horizontal reference line is the line from sella turcica to point N—the line (S-N). In the average individual, the S-N plane is oriented at 6 to 7 degrees upward anteriorly to the Frankfort plane. Another way to obtain a Frankfort line is simply to draw it at a specific inclination to S-N, usually 6 degrees

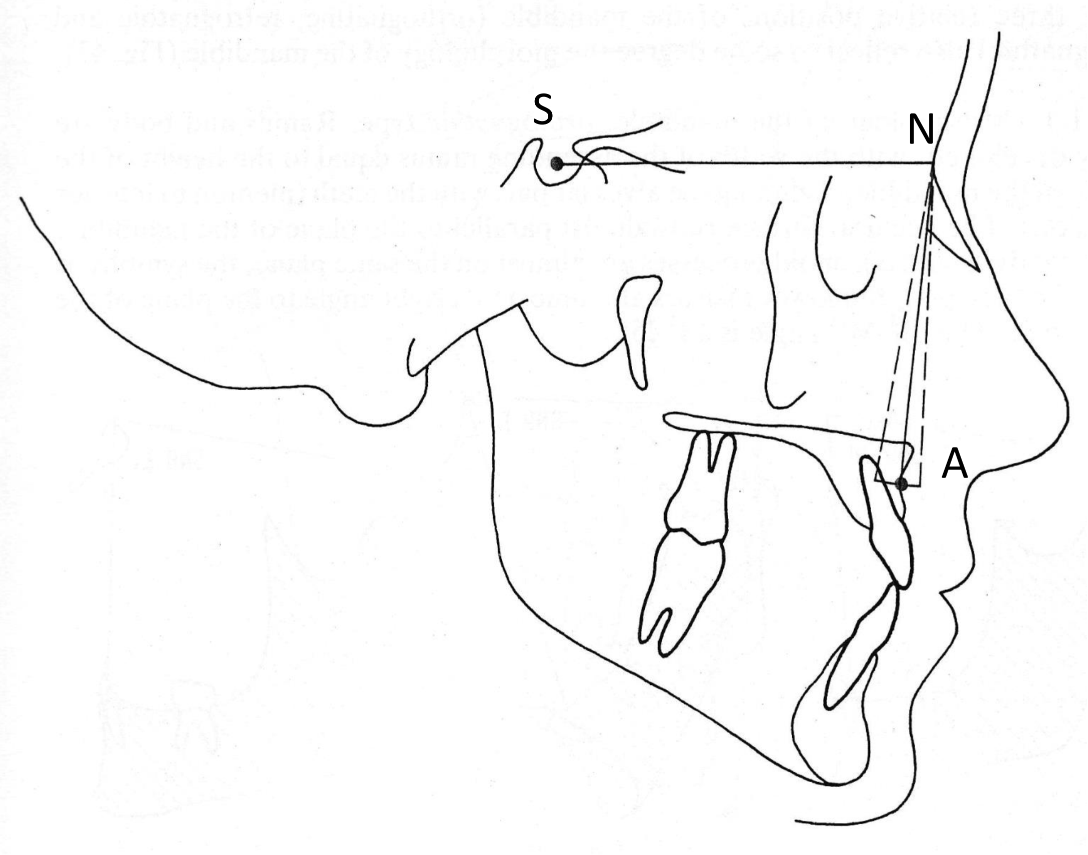


Cephalometric Angles - SNA

- The SNA angle measures the position of the **maxilla** (upper jaw) relative to the anterior cranial base.
- **Formed by:** The intersection of two lines: **S-N** (Sella to Nasion) and **N-A** (Nasion to Point A).
- Normal Value: Approximately 82

Interpretation:

- **Increased SNA (> 84):** Indicates **maxillary protrusion** (the upper jaw is too far forward).
- **Decreased SNA (< 80):** Indicates **maxillary retrusion** (the upper jaw is too far back).

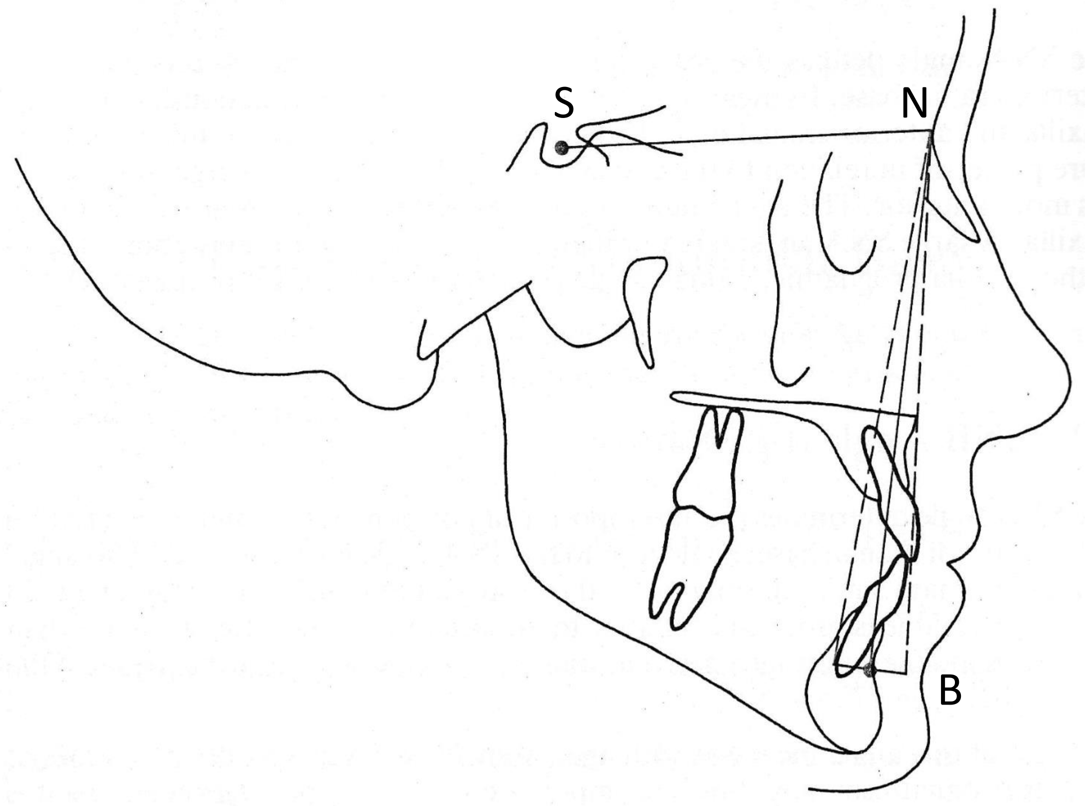


Cephalometric Angles - SNB

- The SNB angle measures the position of the **mandible** (lower jaw) relative to the anterior cranial base.
- **Formed by:** The intersection of two lines: **S-N** (Sella to Nasion) and **N-B** (Nasion to Point B).
- **Normal Value:** Approximately **80**

Interpretation:

- **Increased SNB (> 82):** Indicates **mandibular protrusion** (prognathism/chin is far forward).
- **Decreased SNB (< 78):** Indicates **mandibular retrusion** (retrognathism/chin is receded).

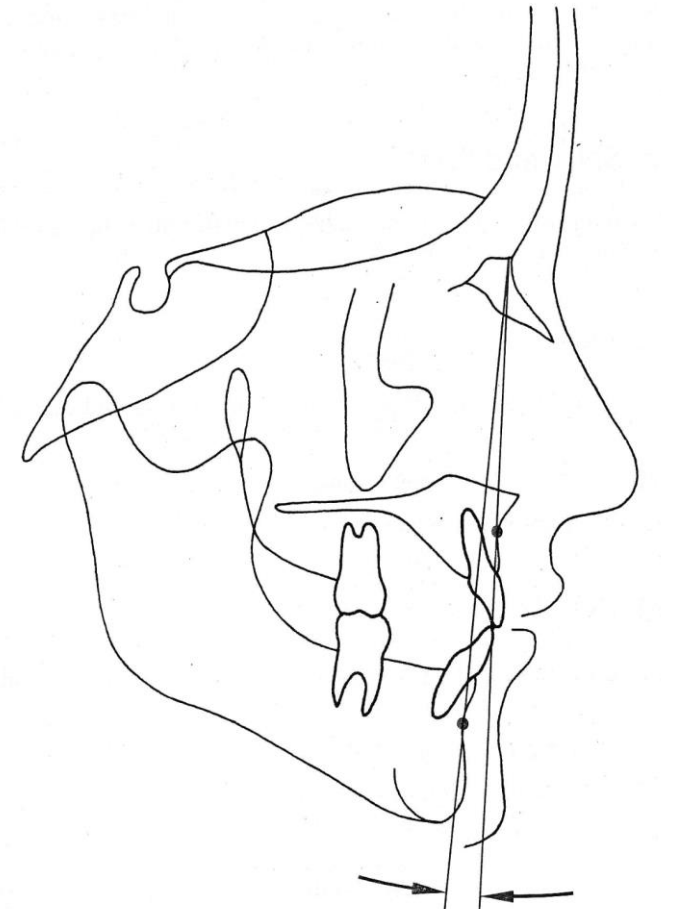


Cephalometric Angles - ANB

- The **ANB angle** is perhaps the most important measurement in Steiner's analysis because it defines the **skeletal relationship** between the upper and lower jaws, regardless of where they sit on the head.
- **Normal Value:** Approximately 2 to 4

Interpretation:

- **Increased ANB (> 4):** Indicates the upper jaw is significantly ahead of the lower jaw. This can be caused by a protrusive maxilla (high SNA), a retrusive mandible (low SNB), or a combination of both.
- **Decreased SNB (< 0):** The lower jaw is ahead of the upper jaw

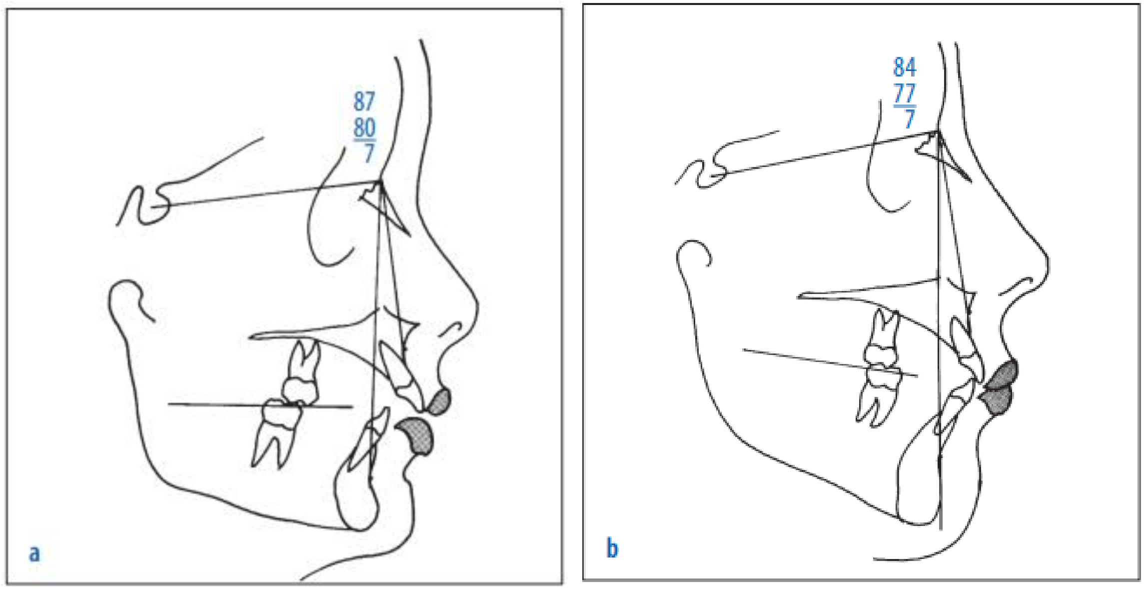


Generative AI Exercise

- ChatGPT
- Gemini
- Copilot
- Claude

Wits Appraisal

- While the ANB angle is popular, it can be unreliable in two situations:
  - Relative position of point N
  - Tilt of dentition



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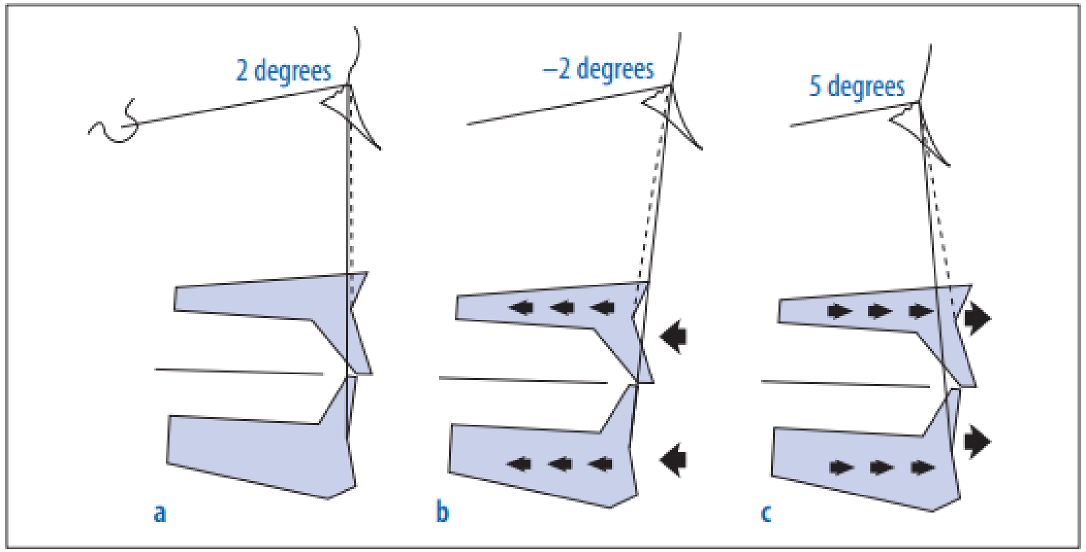

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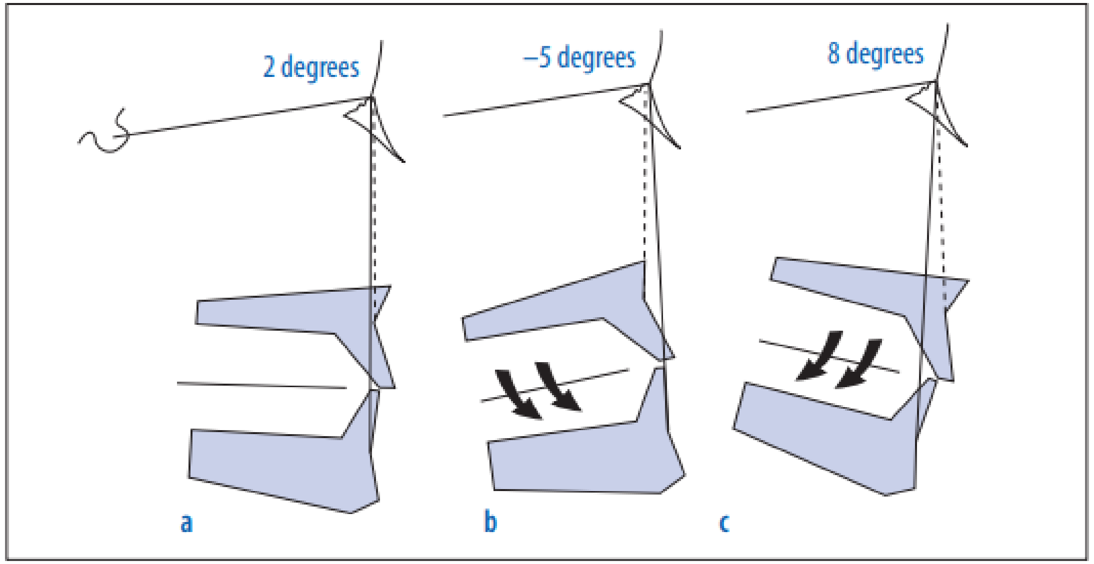
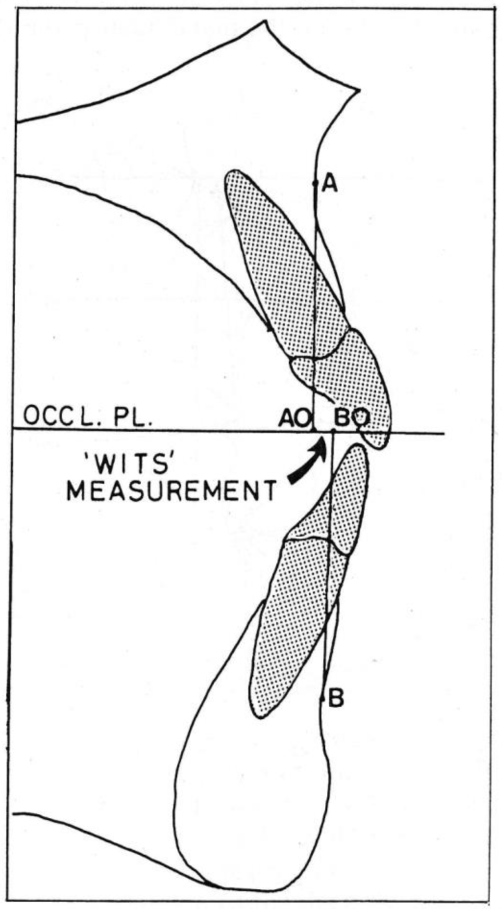


Wits Appraisal

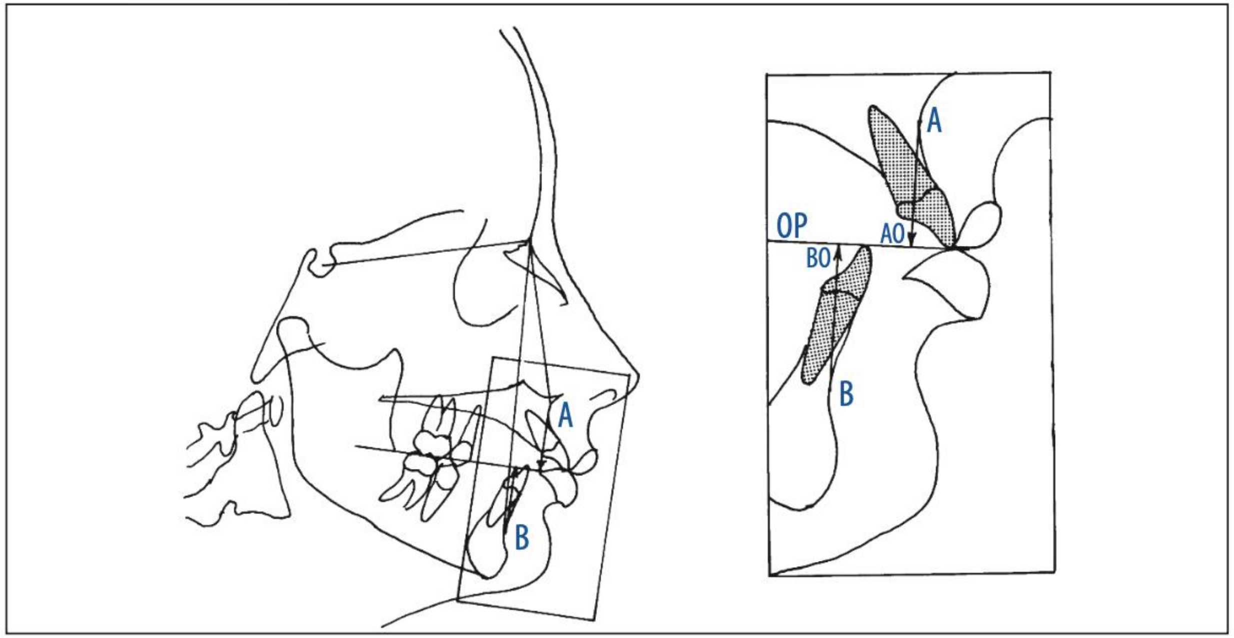
- The Wits Appraisal solves this by relating the jaws only to each other, using the **Functional Occlusal Plane** as a reference



Wits Appraisal

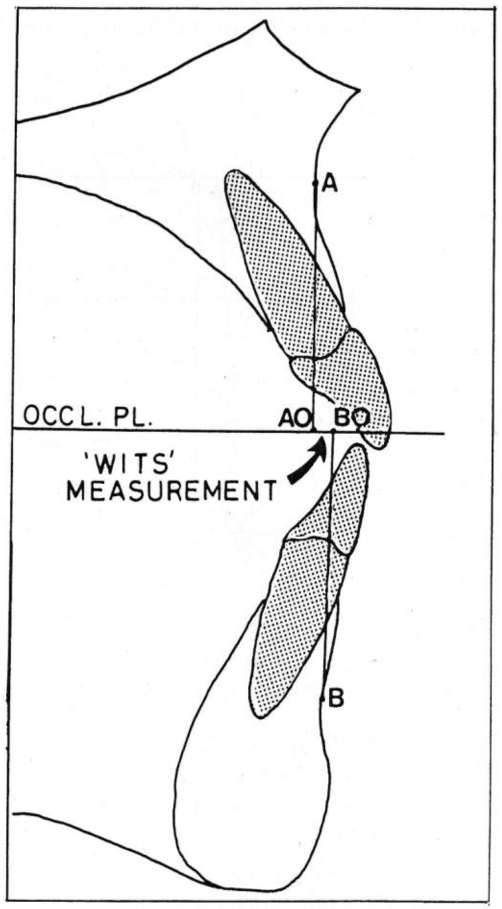
How to Measure Wits Appraisal

1. **Draw the Functional Occlusal Plane:** Draw a line through the maximum intercuspation of the posterior teeth (premolars and molars).
2. **Drop Perpendiculars:** From Point **A** (maxilla) and Point **B** (mandible), draw lines straight down (perpendicular) to the occlusal plane.
3. **Label the Points:** Mark the spots where these lines hit the plane as **AO** and **BO**.
4. **Measure the Distance:** Measure the distance between AO and BO in millimeters.



Wits Appraisal

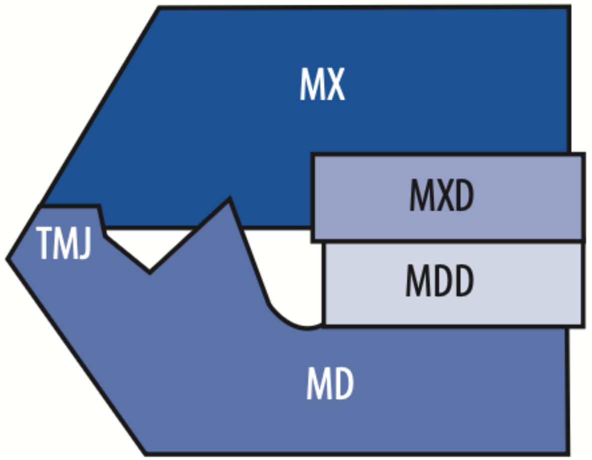
- The normal **Wits Appraisal** value typically centers around 0 mm, indicating ideal jaw alignment (Class I), with ranges like -1 mm to +1 mm often cited as normal, though values vary by gender, ethnicity, and study, generally with males often slightly negative (around -1 mm) and females near 0 mm or slightly positive



Analysis (Year)	Key Clinical Purpose	Important Measurements & Norms
Downs (1948)	First systematic analysis of facial skeletal patterns and dental correlation.	**MP/SN Angle:** Determines facial divergence (27–37°). **Bolton Triangle:** Uses Nasion, Sella, and Bolton point for growth.
Steiner (1953)	Relates skeletal and dental factors to create a treatment plan.	**SNA/SNB/ANB:** Maxilla/Mandible relation to cranial base. **Incisor to N-A/N-B:** Linear distance (4mm) and axial inclination.
Tweed (1953)	Focused on lower incisor stability and the "Tweed Triangle."	**FMA:** Frankfort-Mandibular Plane Angle (20–28°). **IMPA:** Lower incisor to mandibular plane (90°).
Jarabak (1972)	Uses a facial polygon to predict growth patterns (CW vs. CCW).	**Jarabak Ratio:** Posterior facial height / Anterior facial height. **Sum of Angles:** Saddle + Articular + Gonial angles.
Wits (1975)	Purely linear evaluation of the A-P jaw relationship.	**AO-BO Distance:** Perpendicular distance between Points A and B on the occlusal plane
McNamara (1984)	Relates jaws to the Nasion-perpendicular; ideal for surgery.	**N-Perp to Point A:** Maxillary position. **N-Perp to Pogonion:** Mandibular position.

McNamara Analysis

1. Maxilla to cranial base
   - Soft tissue evaluation
   - Hard tissue evaluation
2. Maxilla to mandible
   Anteroposterior relationship
   Vertical relationship
3. Mandible to cranial base
4. Dentition



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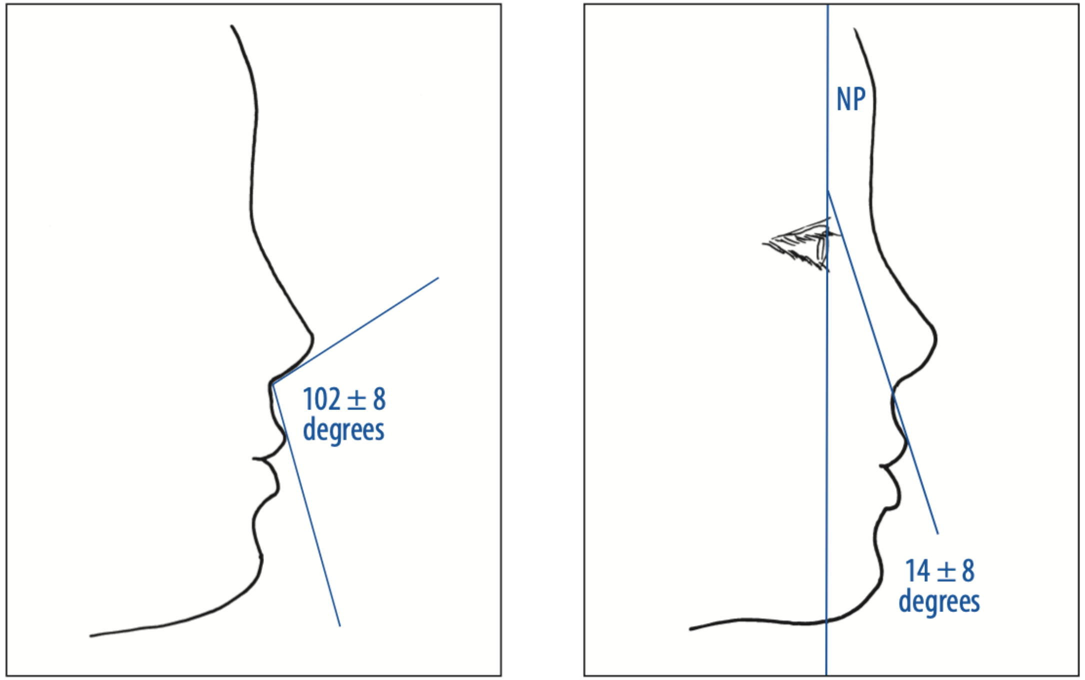
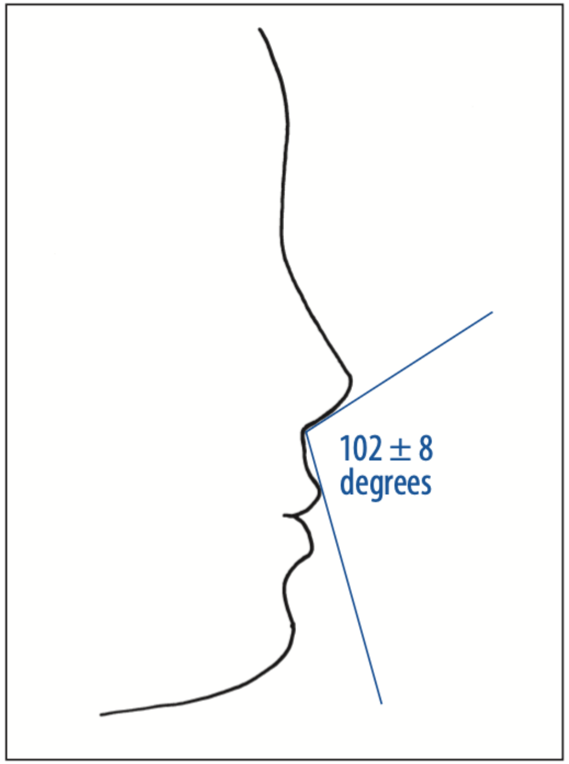


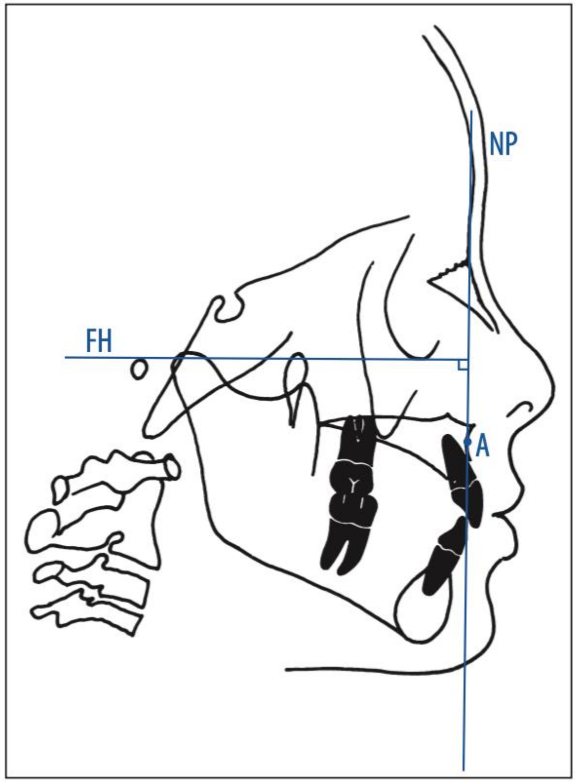
McNamara Analysis- Soft Tissue Evaluation

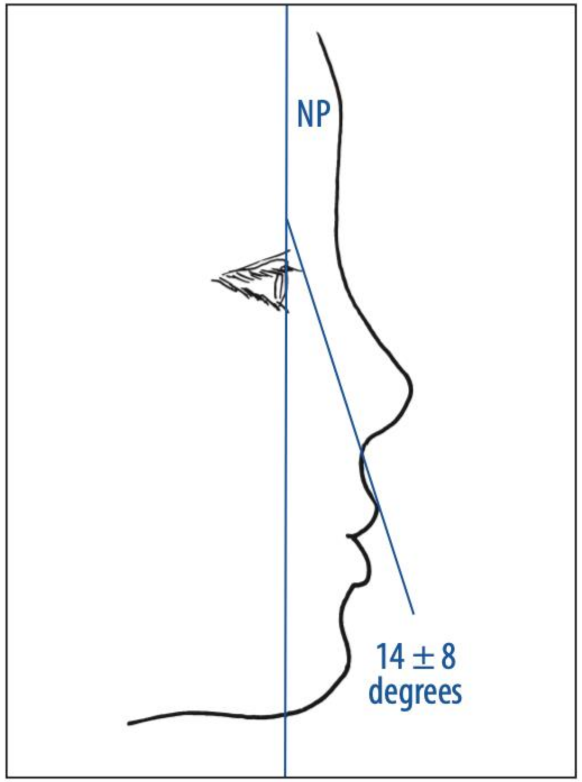
- The **Nasolabial angle** is formed by drawing a line tangent to the base of the nose and a line tangent to the upper lip



McNamara Analysis- Soft Tissue Evaluation

- The **Cant of the Upper Lip** is evaluated by constructing an angle using a line tangent to the upper lip and the nasion-perpendicular. The nasion-perpendicular is a vertical line drawn perpendicular to Frankfort horizontal (FH) through nasion. Note that FH extends from the superior aspect of the external auditory meatus [anatomic porion]





McNamara Analysis

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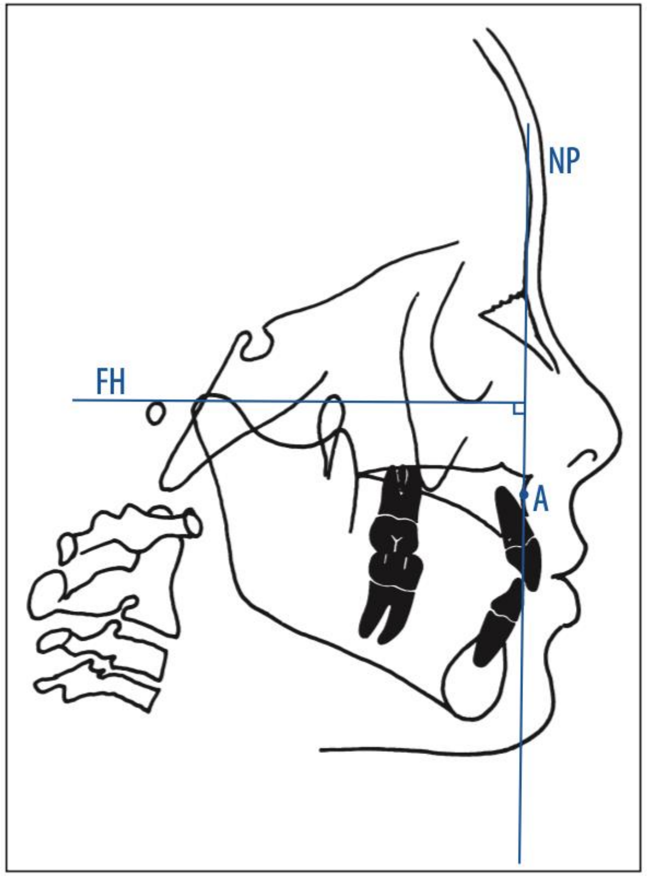
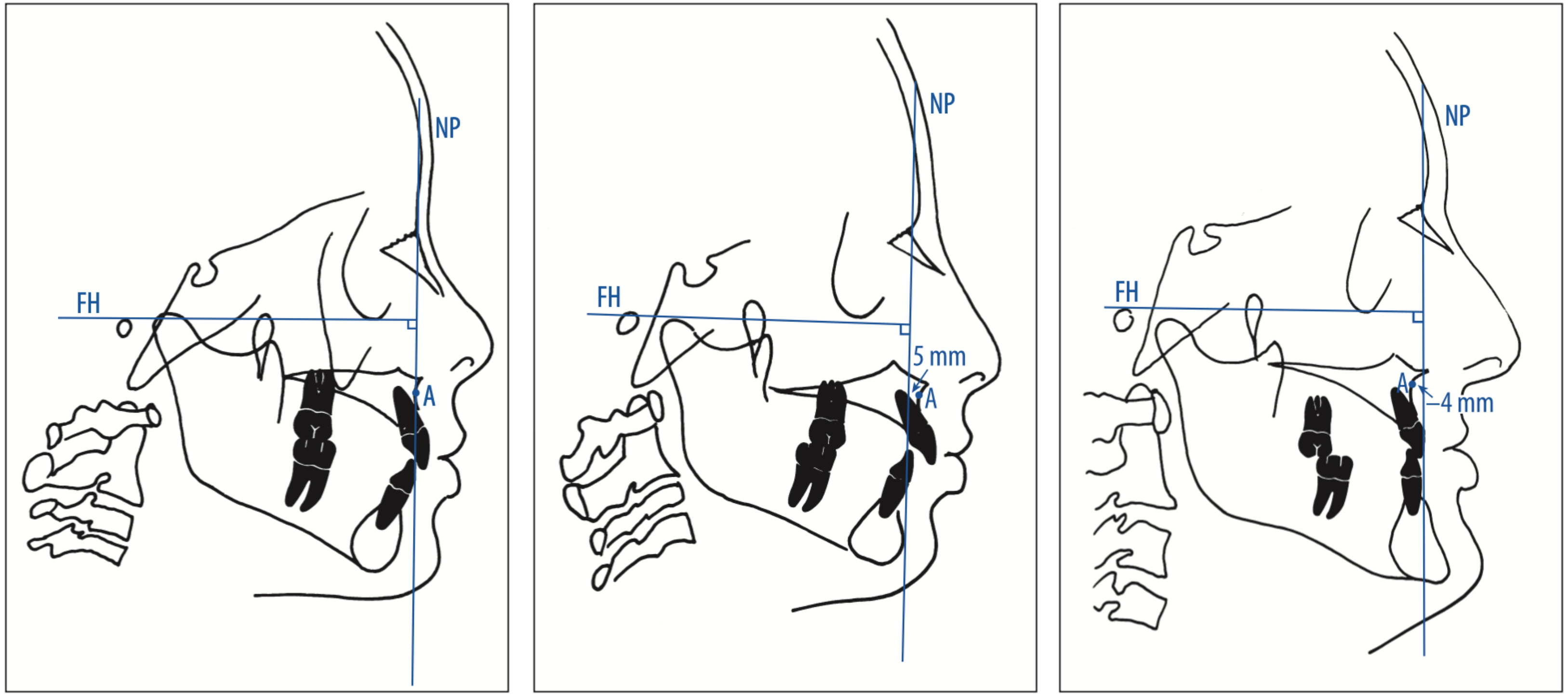


In a well-balanced face, nasion-perpendicular (NP) is within 1 mm of point A.


NP
FH
A
NP
FH
5 mm
A
NP
FH
A
−4 mm

McNamara Analysis

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   - Hard tissue evaluation
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   Anteroposterior relationship

   Vertical relationship
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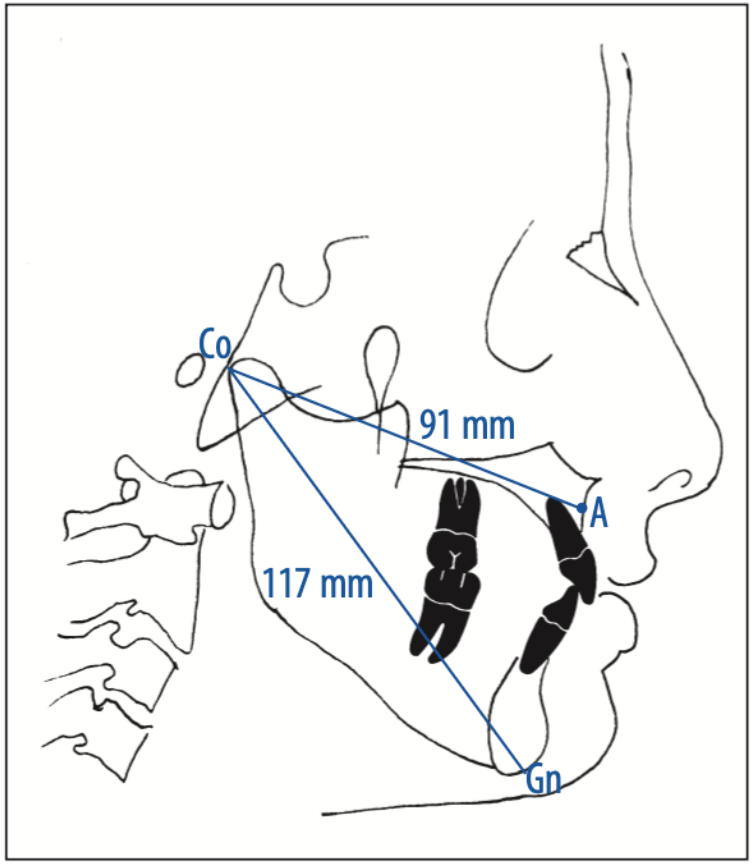
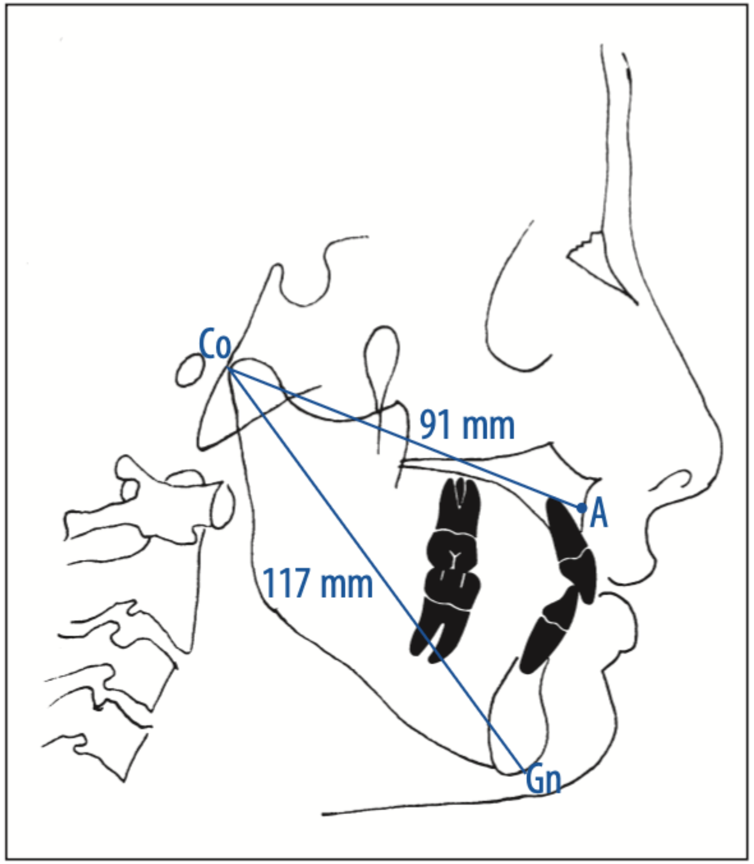


McNamara Analysis- Anteroposterior Relationship

- Midfacial length is measured from **Condylion** to **point A**. The effective length of the mandible is measured from **Condylion** to anatomic **Gnathion**



McNamara Analysis- Anteroposterior Relationship

Table 10-1 Normative Standards in McNamara Analysis

Midfacial length (mm) (Co-A)	Mandibular length (mm) (Co-Gn)	Lower anterior facial height (mm) (ANS-Me)
80	97–100	57–58
81	99–102	57–58
82	101–104	58–59
83	103–106	58–59
84	104–107	59–60
85	105–108	60–62
86	107–110	60–62
87	109–112	61–63
88	111–114	61–63
89	112–115	62–64
90	113–116	63–64
91	115–118	63–64
92	117–120	64–65
93	119–122	65–66
94	121–124	66–67
95	122–125	67–69
96	124–127	67–69
97	126–129	68–70
98	128–131	68–70
99	129–132	69–71
100	130–133	70–74
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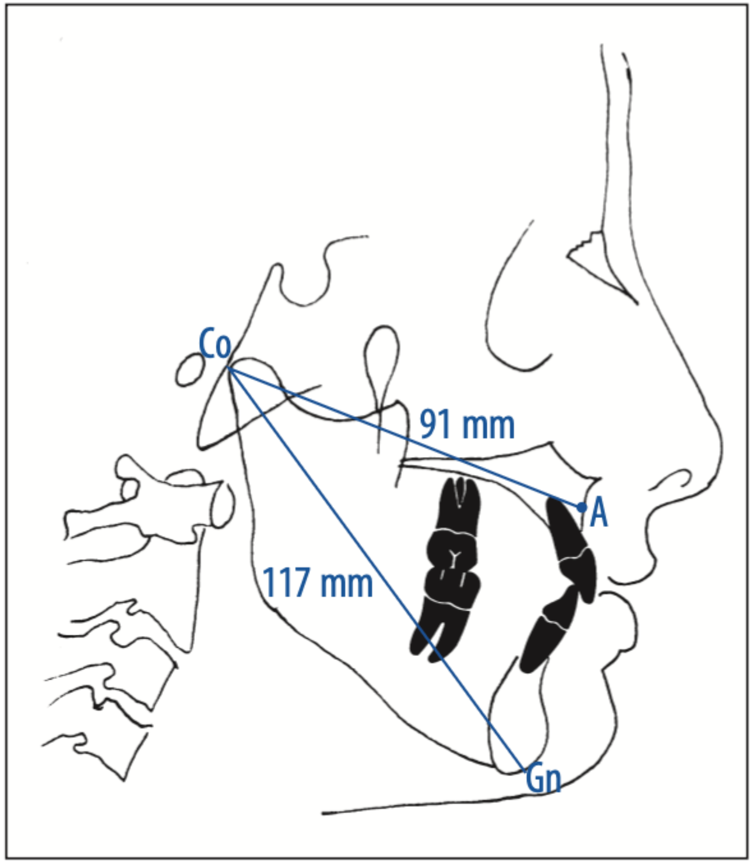
104	137–140	74–78
105	138–141	75–79



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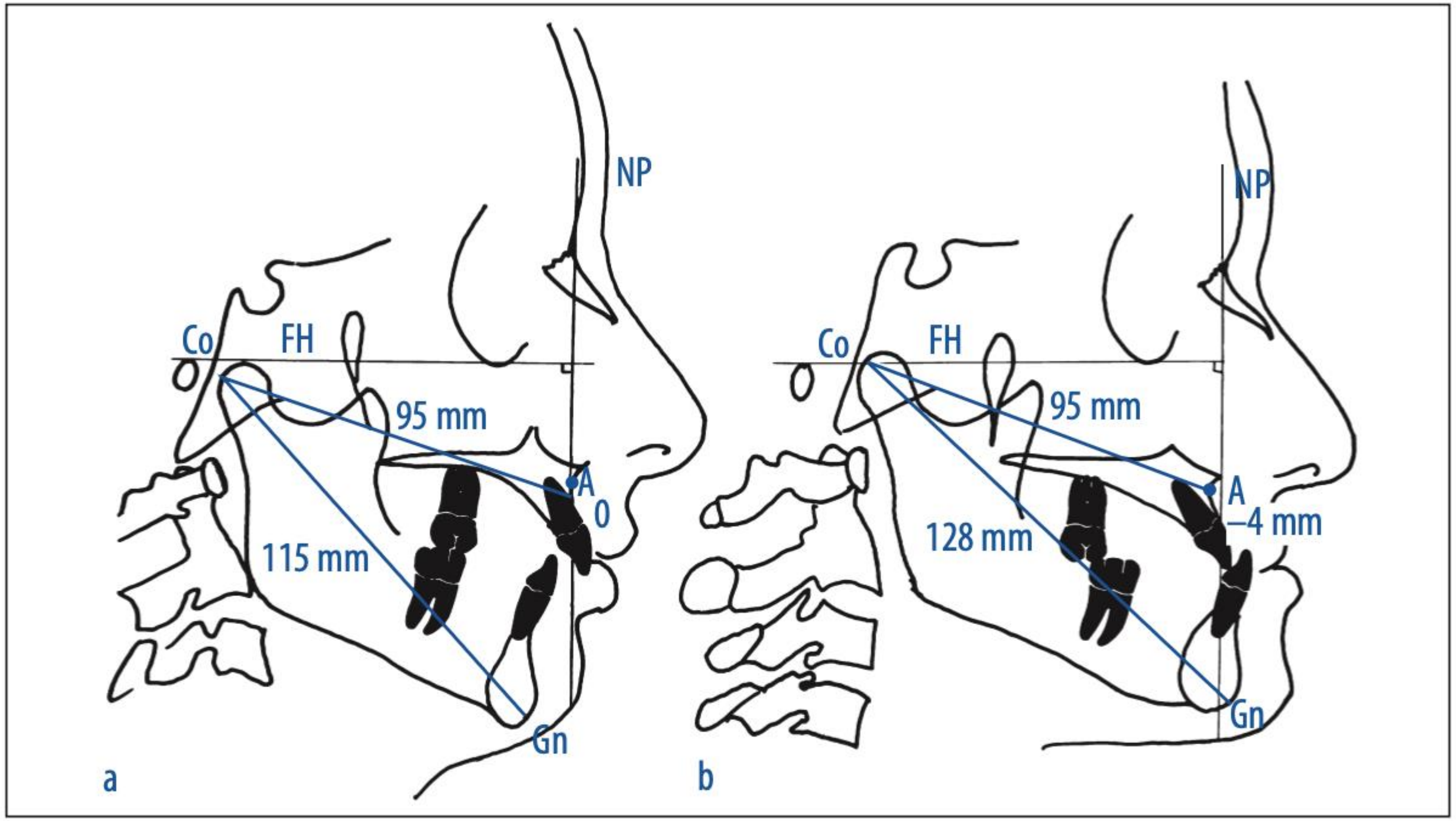
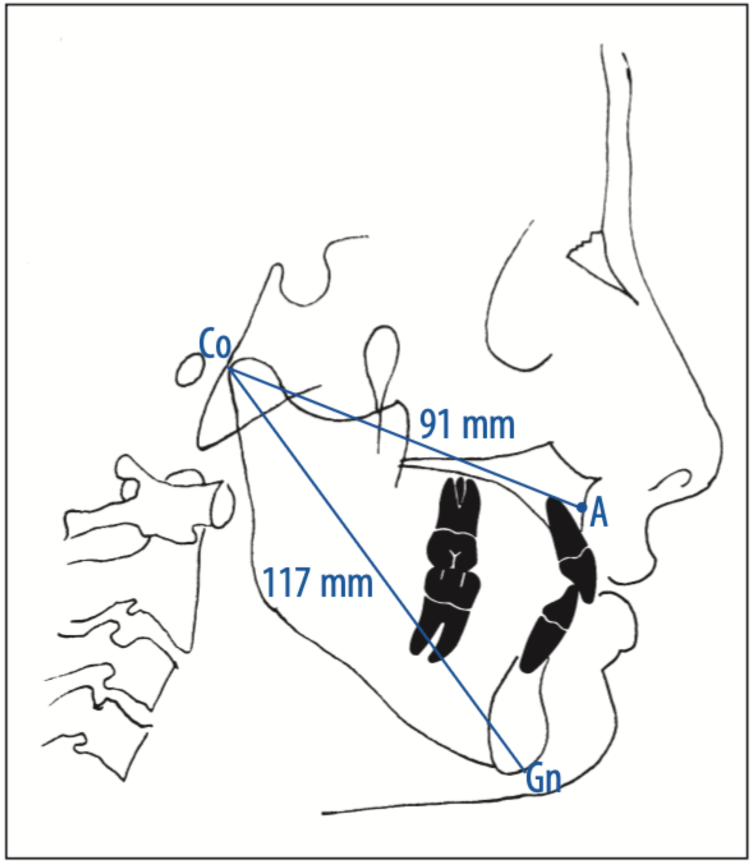
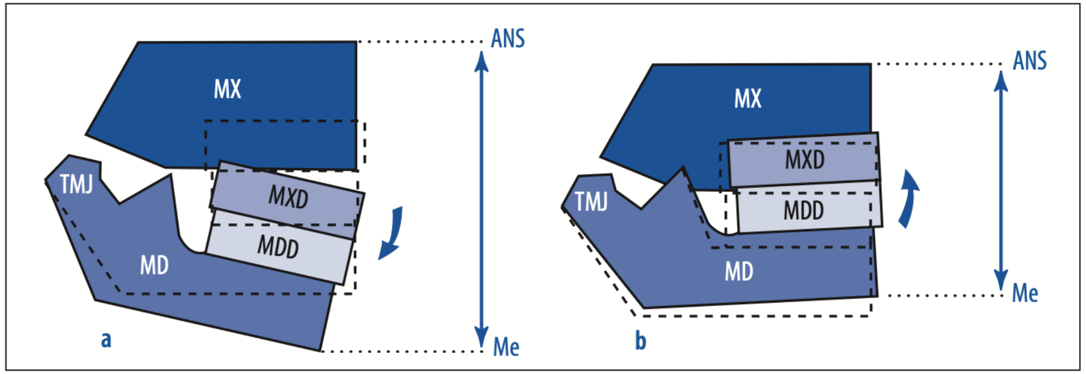
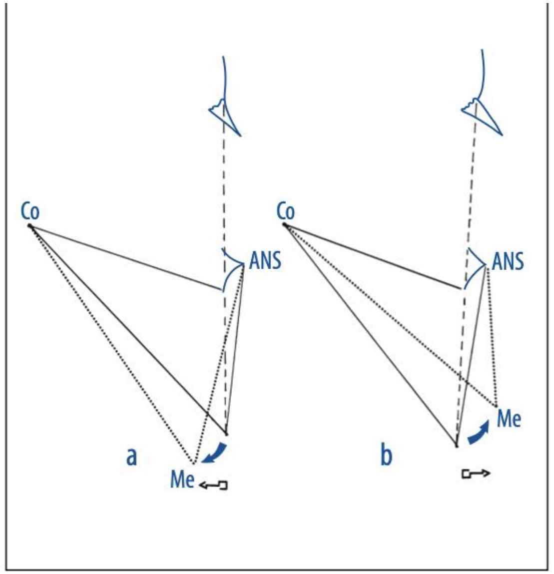




McNamara Analysis- **Vertical relationship**

- Vertical maxillary excess can cause a downward and back-ward rotation of the mandible, resulting in an increase in lower anterior facial height (LAFH)
- Conversely, vertical maxillary deficiency will cause the mandible to rotate upward and forward, thereby reducing the LAFH





McNamara Analysis- Vertical relationship

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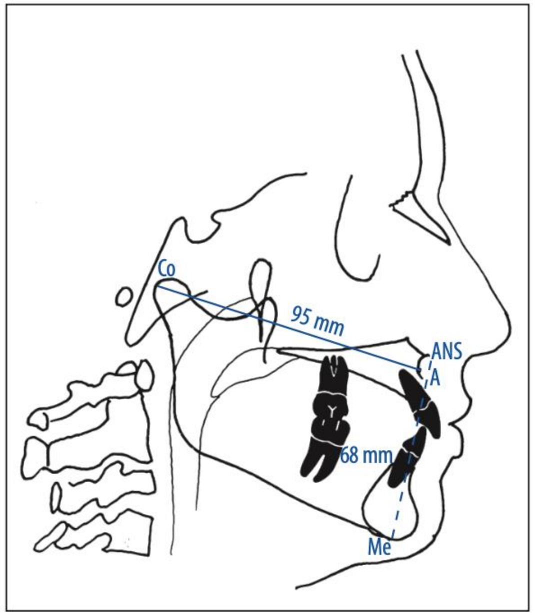
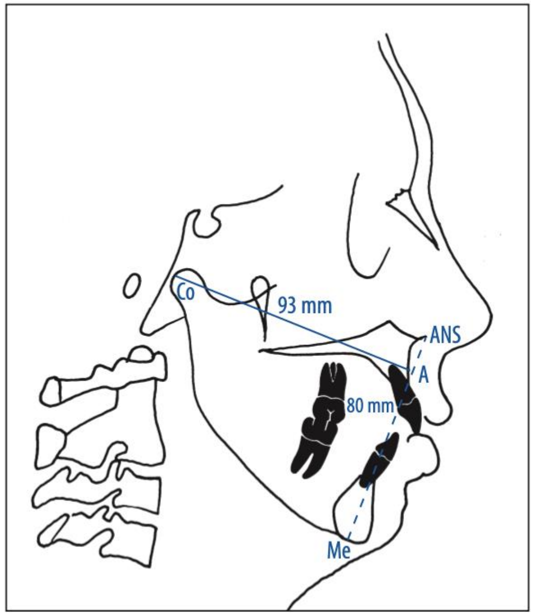
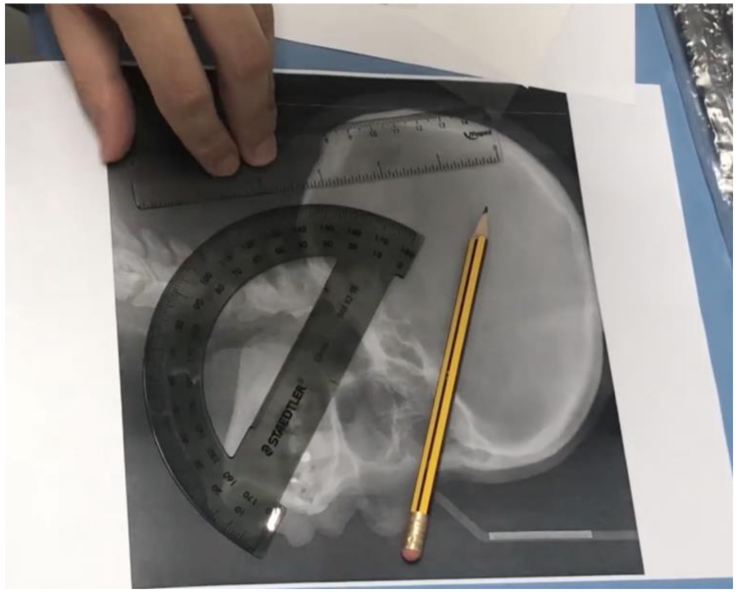
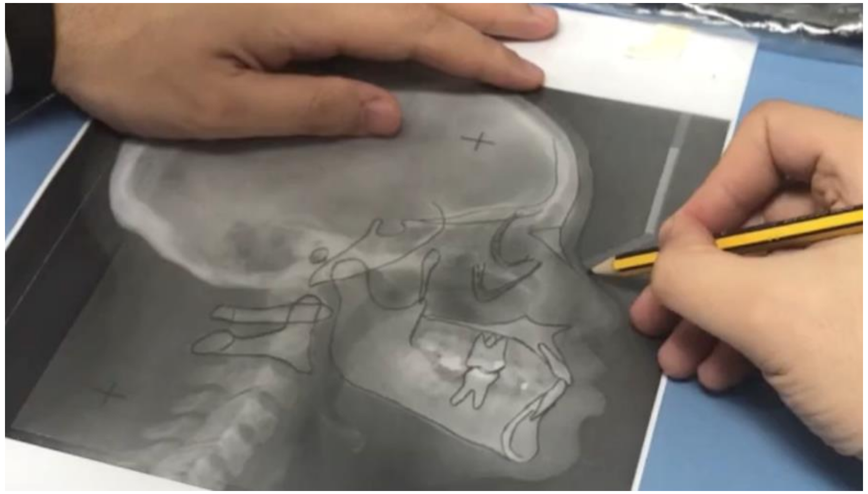




Transferring Cephalometric Data to diagnostic information

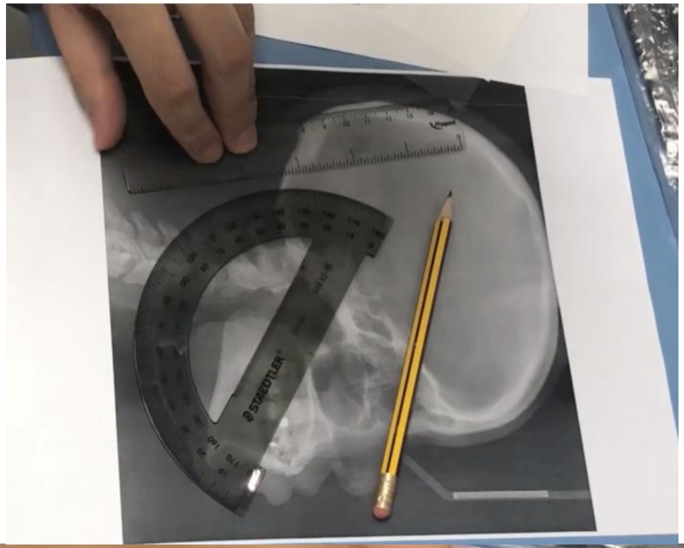
- In the "pre-digital" era of dentistry, transferring a raw lateral cephalogram into diagnostic data was a meticulous, manual process that required specialized physical tools and about 20–30 minutes of clinical time. A sheet of clear **matte acetate tracing paper** was taped directly over the X-ray film





Transferring Cephalometric Data to diagnostic information

- The Physical Setup
- Manual Anatomical Tracing
- Landmark Identification
- Construction of Planes
- The Measurement Phase
  - Angular Measurements
  - Linear Measurements
  - Manual Comparison

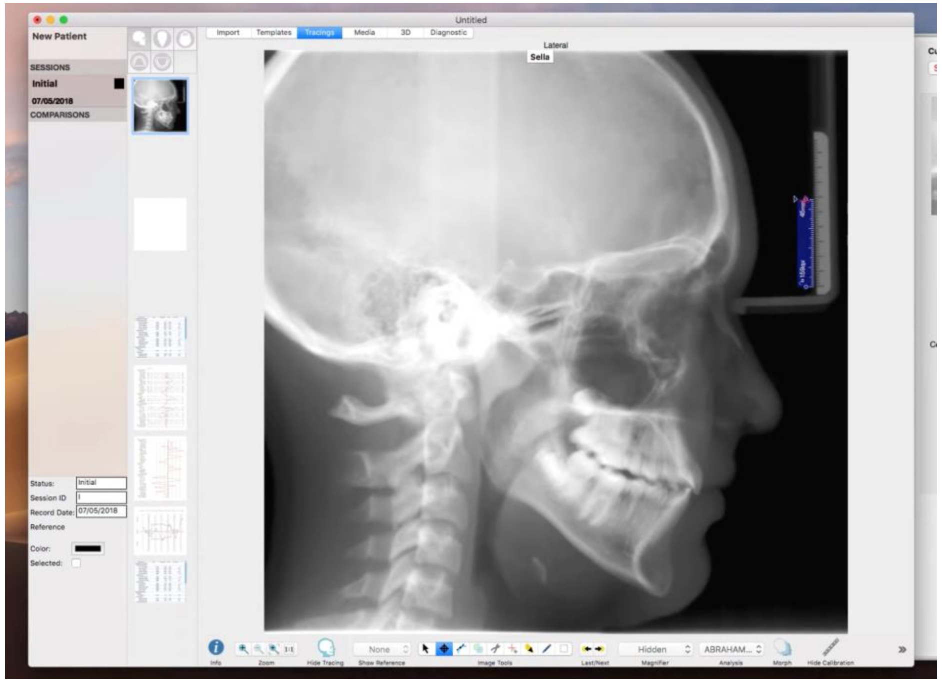


Transferring Cephalometric Data to diagnostic information- **Computerized Workflow**

- Unlike manual tracing on acetate paper, the software uses pixels and mathematical algorithms to ensure precision.
- The transition from a raw radiograph to a completed analysis happens in three main stages:

- Standardization & Calibration
- Landmark Detection
  - Manual Digital
  - Semi-Automatic
  - Fully Automatic
- Mathematical Analysis

Transferring Cephalometric Data to diagnostic information- **Computerized Workflow**



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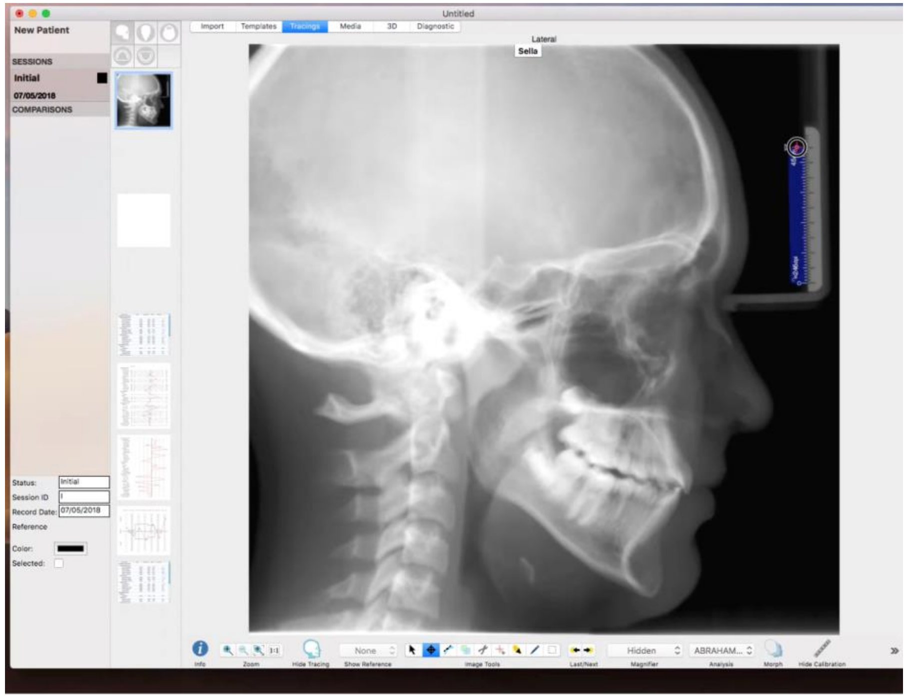

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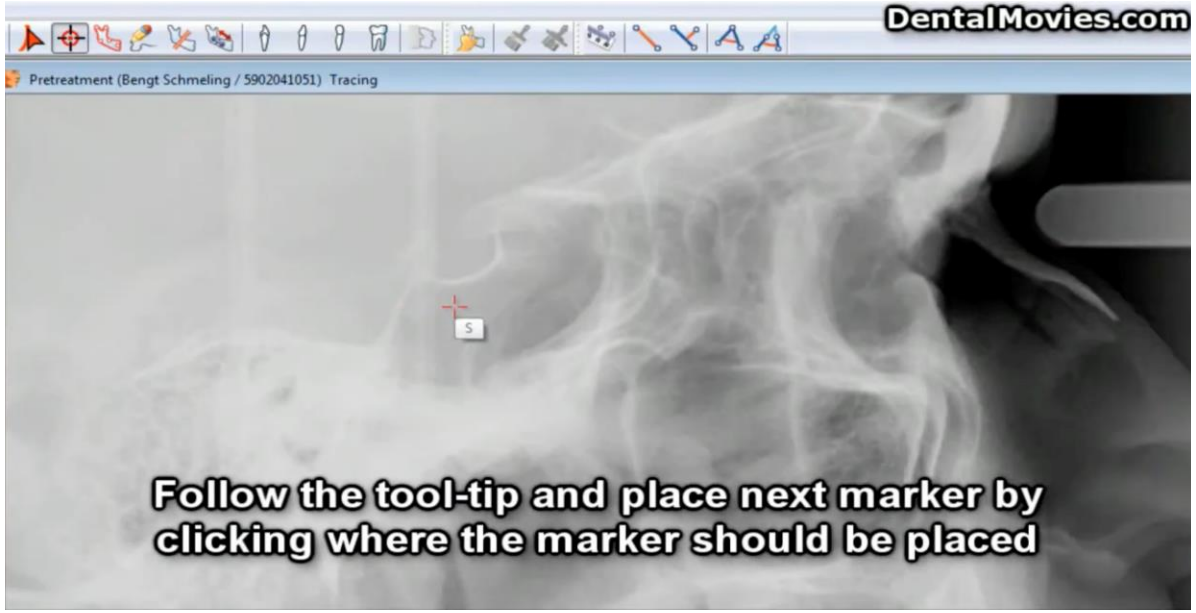

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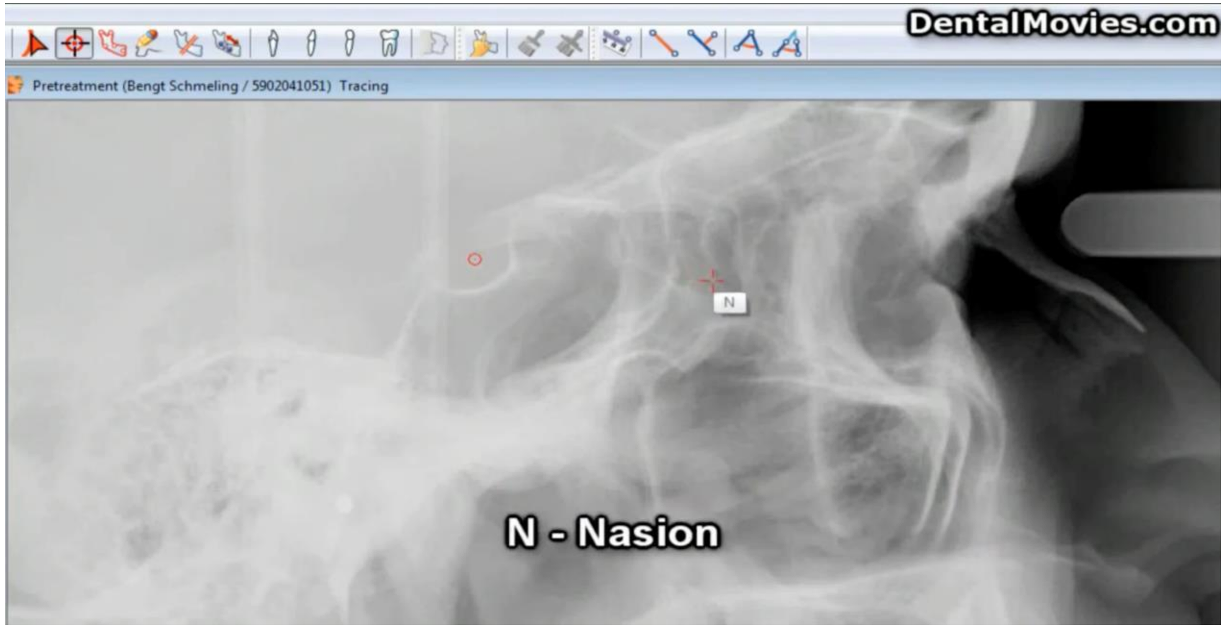

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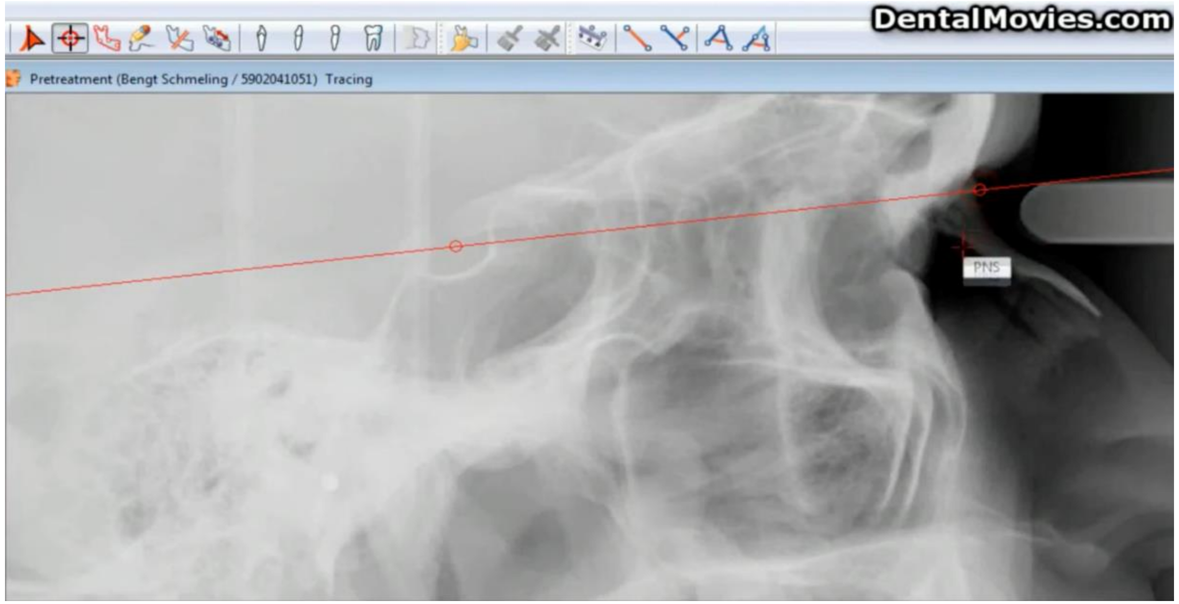

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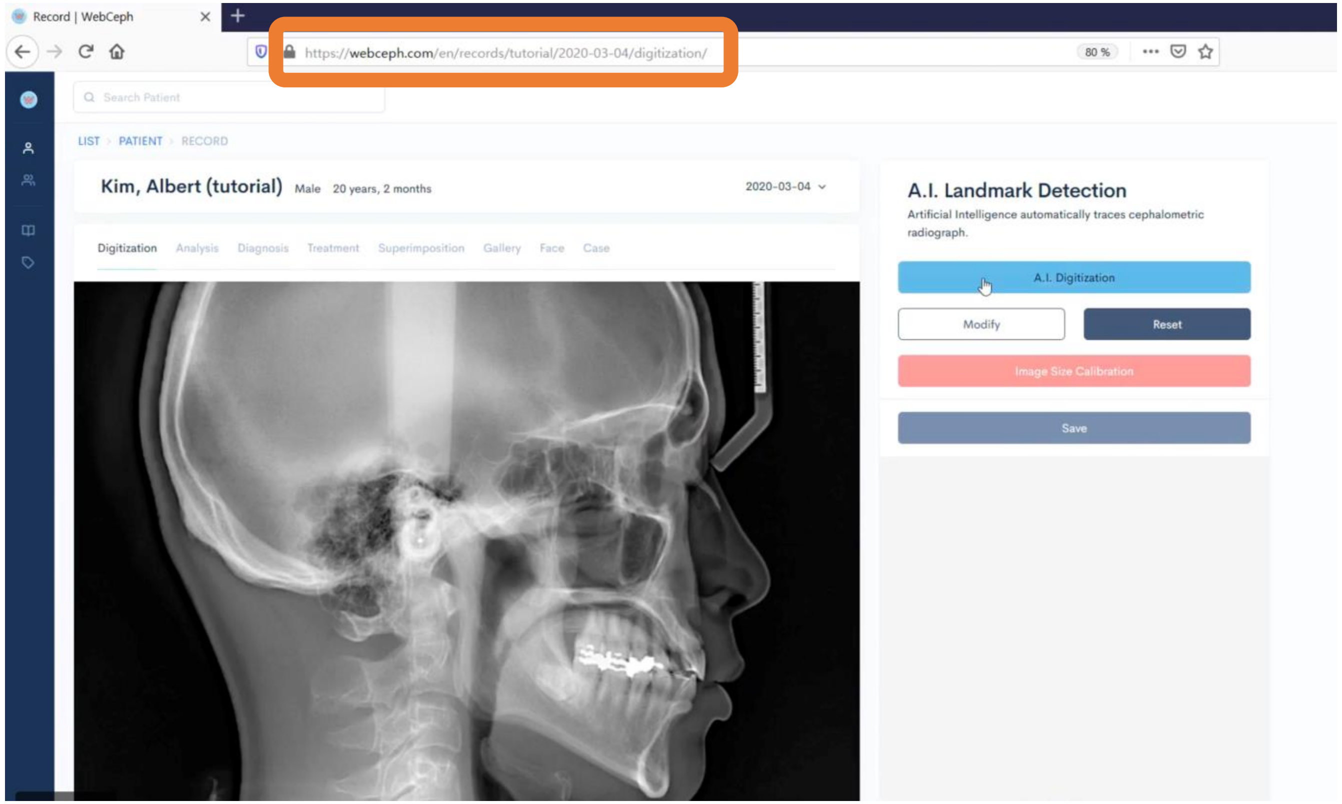

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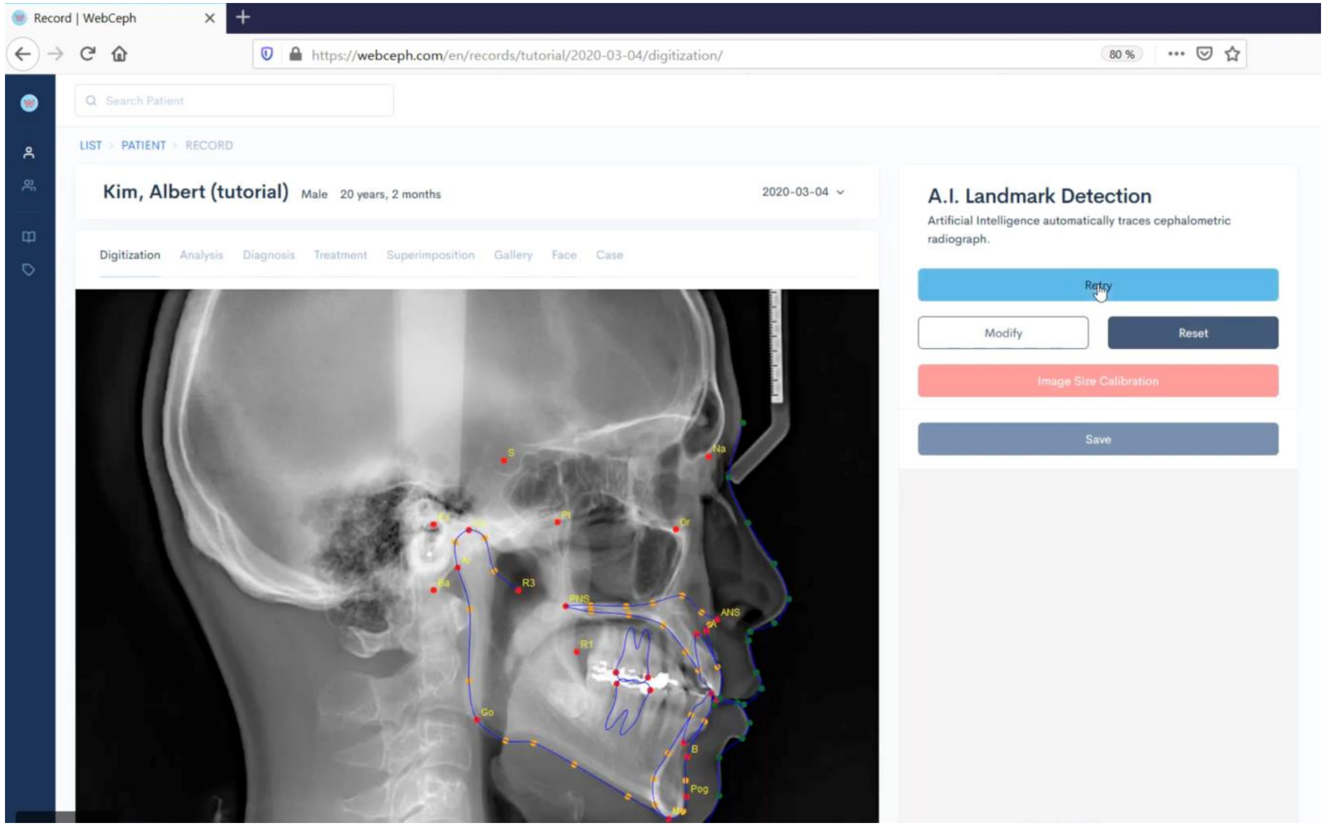

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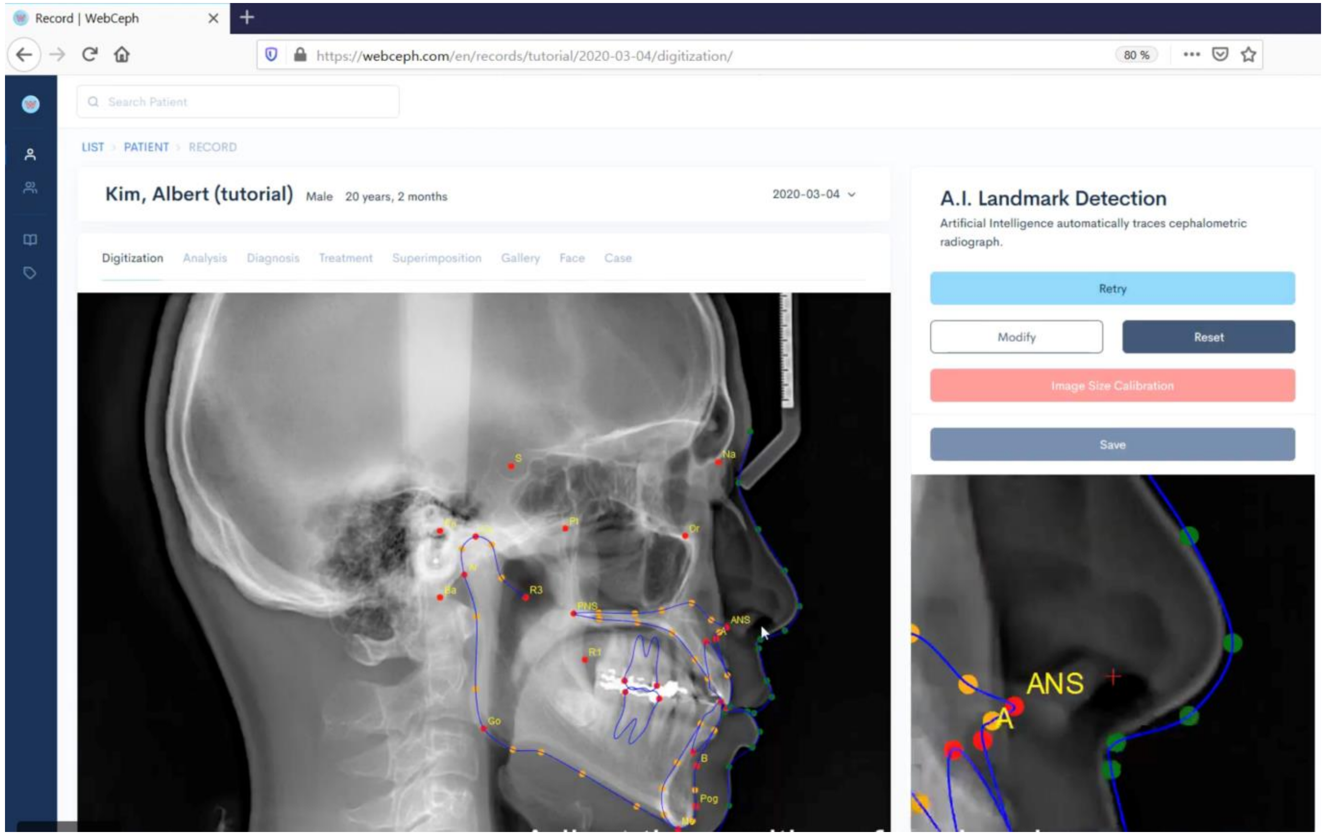

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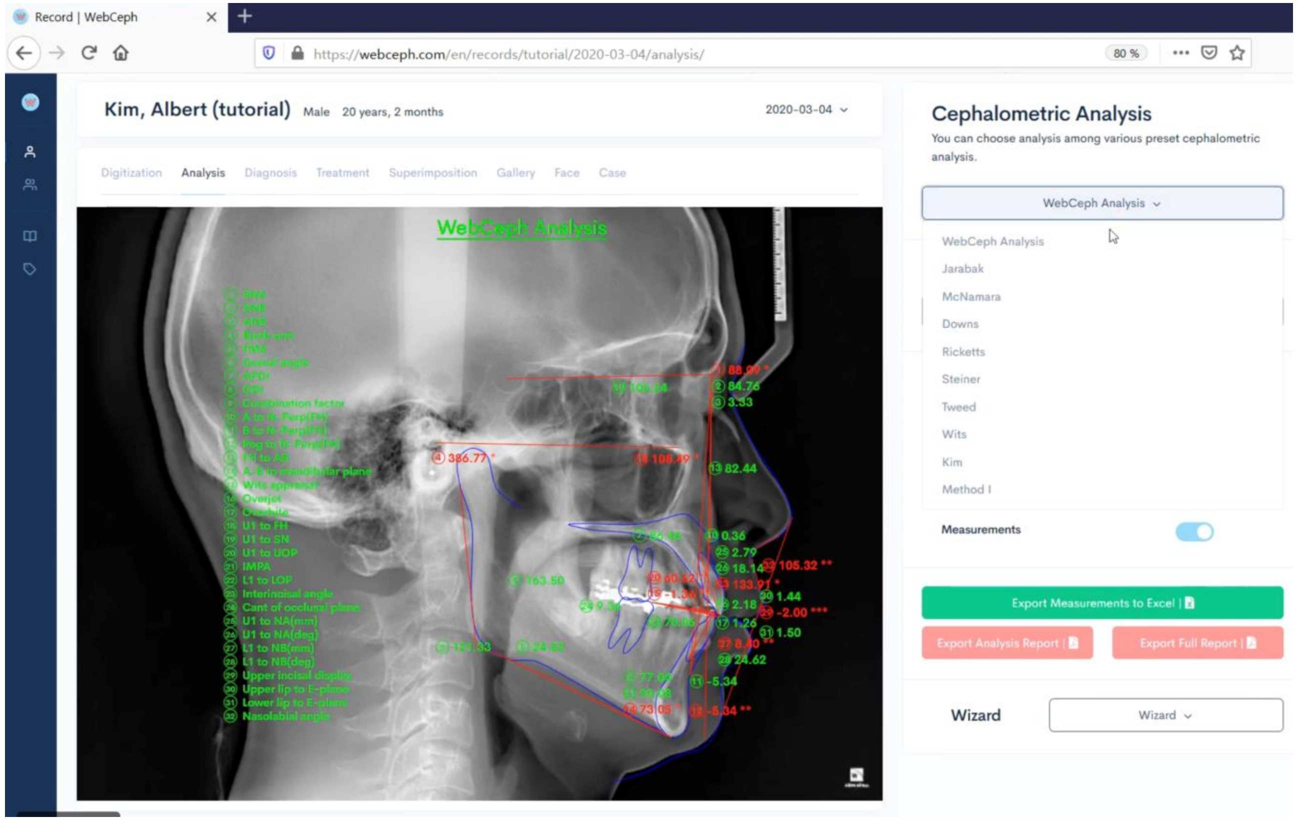

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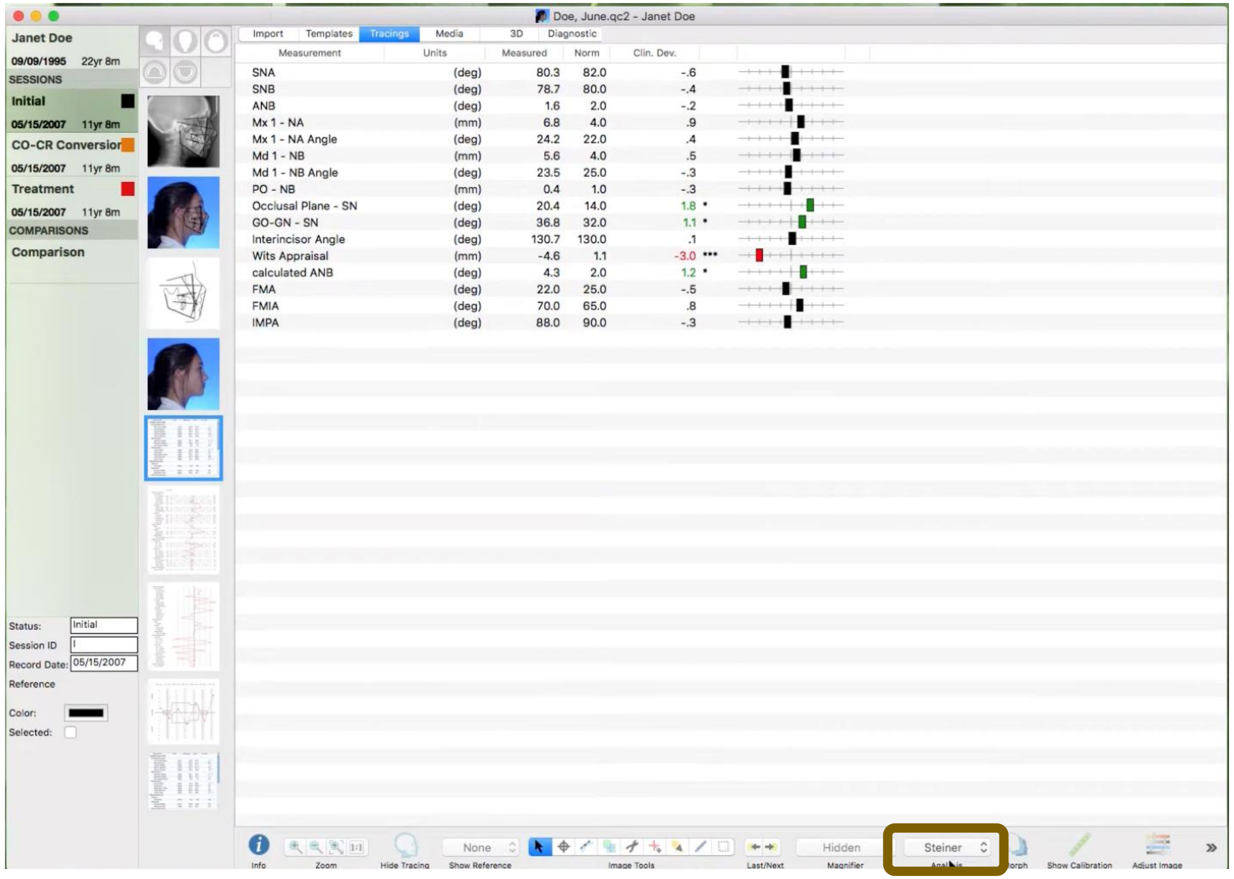

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Cephalometry Role in Dentistry

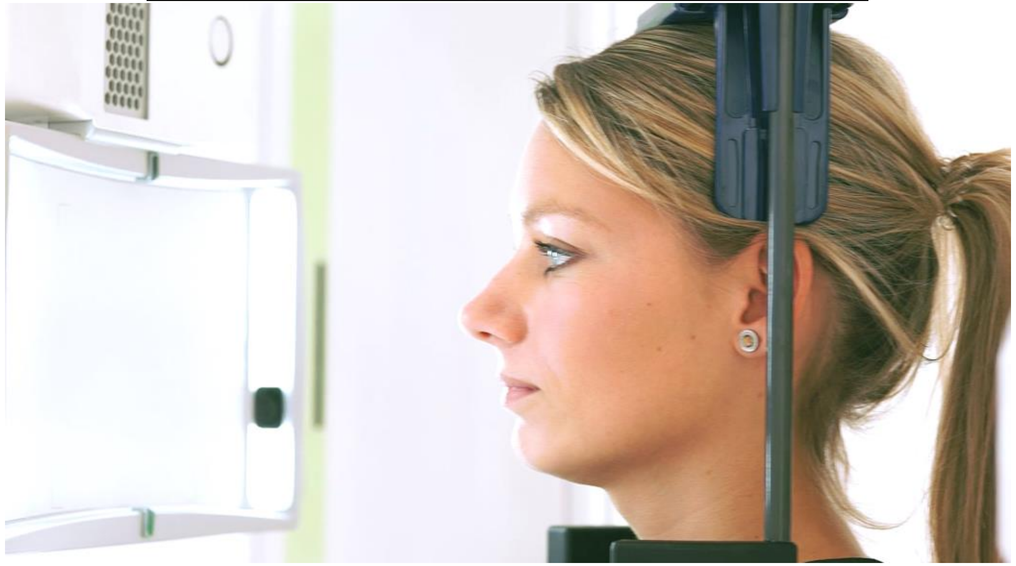
- Cephalometry is no longer the destination — it is one tool inside a soft-tissue–driven, 3D diagnostic process.
- Ceph helps us understand relationships
- Ceph does not tell us the treatment
- Soft tissue is what patients see. Hard tissue is what we move.

Soft-Tissue Paradigm Shift

- **The face is the blueprint, the cephalogram is the limit.”**
- Modern clinicians use the soft tissue to decide *where* they want to go, and the cephalogram to see if the *bone allows* them to get there safely.
-

Virtual Patient

- A virtual patient is a 3D digital representation of the real patient, created by merging three datasets:
  1. Intraoral scan (IOS)
     → Accurate teeth and occlusion
  2. CBCT scan
     → Skeletal structures, roots, airway, TMJ
  3. Facial scan (3D photo)
     → Soft-tissue face in natural head position



Virtual Patient

- 3D face photo together with the CBCT volume are taken and automatically combined in the new generation of x-ray machines –Planmeca and KaVo- imaging software -Romexis- to create a virtual patient

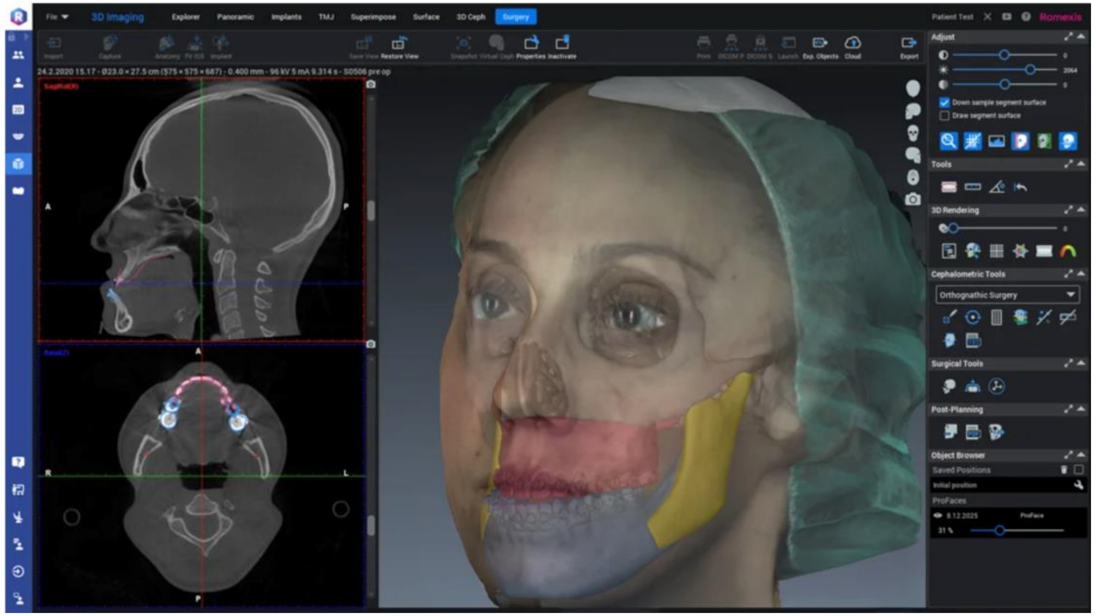
Virtual Patient

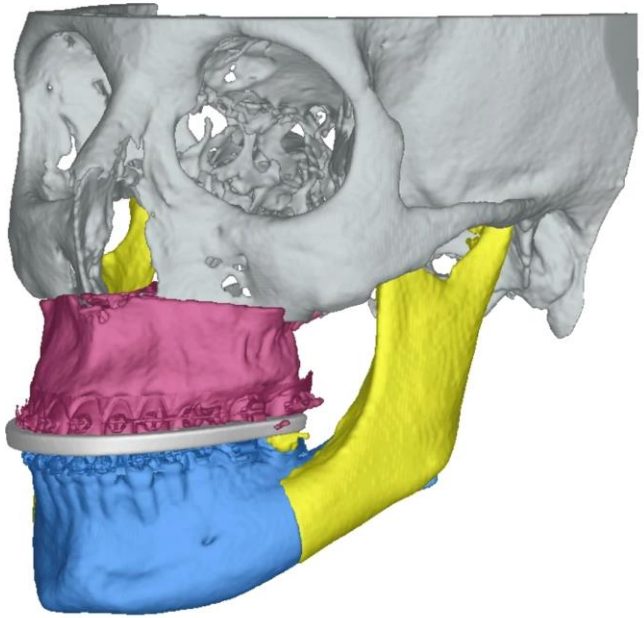
When these three files are **registered and aligned**, you can see:

- Teeth
- Roots
- Jaws
- Face

all in the same coordinate system

- This allows you to analyze **form, function, and esthetics together**, which is impossible with traditional 2D records alone.





Comments? Questions?